AutoCAD

By Dr Mengbi Li



Introduction to AutoCAD



NBD1101 BUILDING DESIGN DOCUMENTATION

By Mengbi Li

Checklist of the CAD practice

- Draw out all your answers (expect practice 1 in the question sheet. Practice 1 is a bonus question)
- Set different line styles
- Use text to identify the questions
- Use XLine function set axis
- Use the function of layers (put objects into different layers. Lock the layer of axis)
- Use the function of dimension (set a new dimension style and apply it)

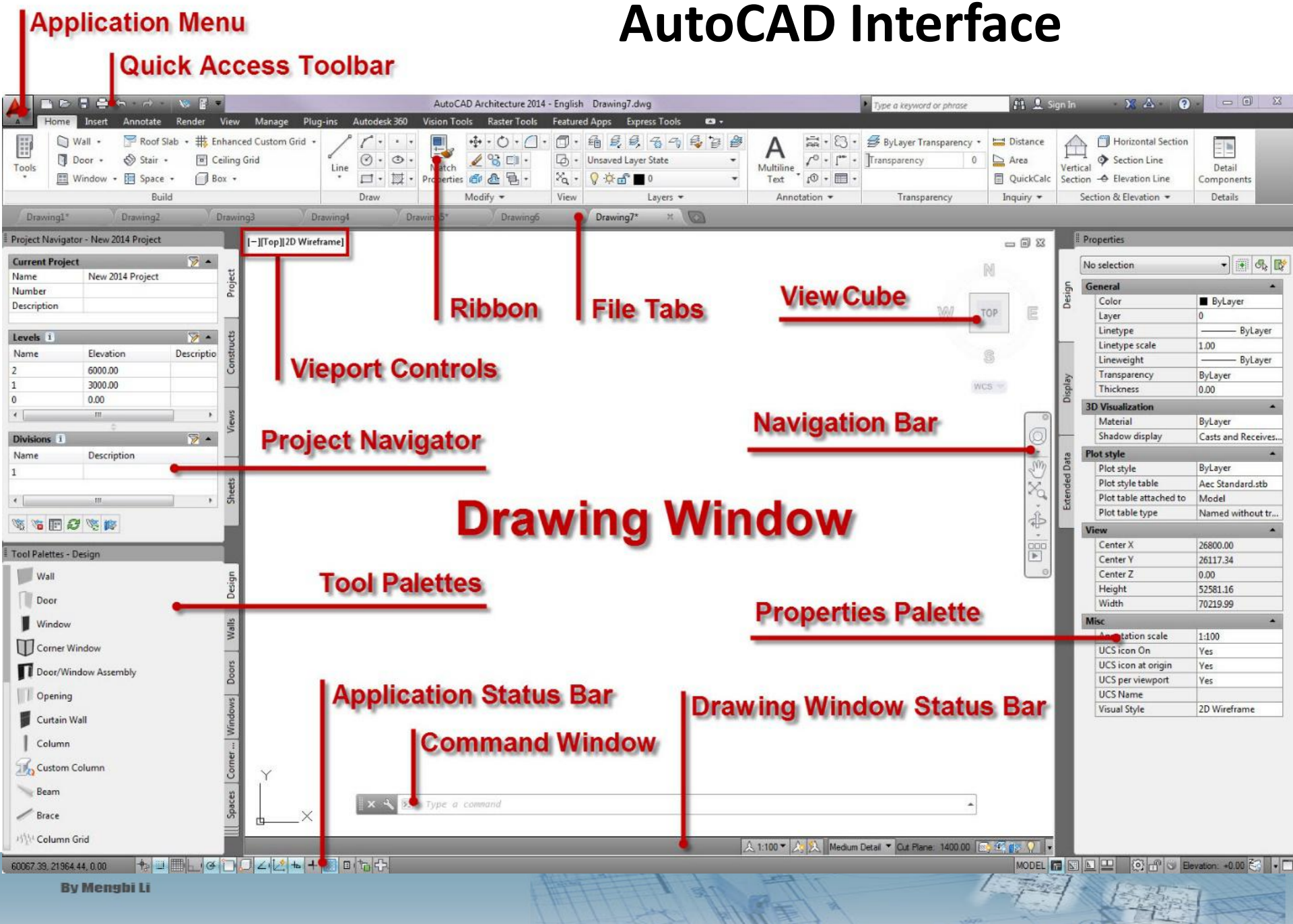


Checklist of the CAD practice

- Create a modified open area by applying blocks
- Create toilets



AutoCAD Interface



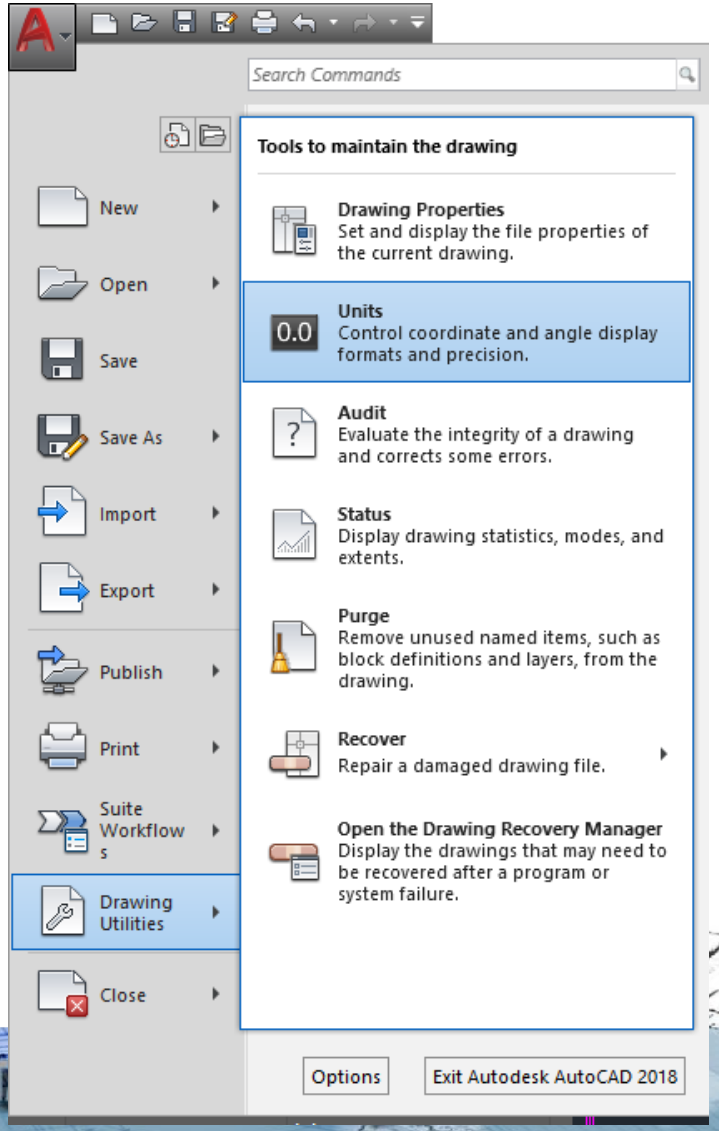
By Mengbi Li

Unit

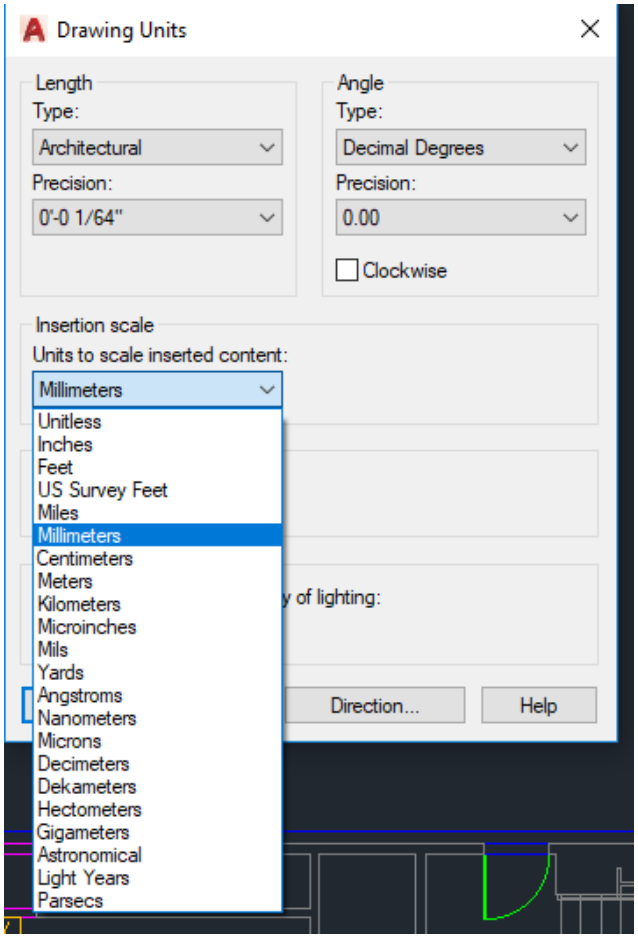
Click 

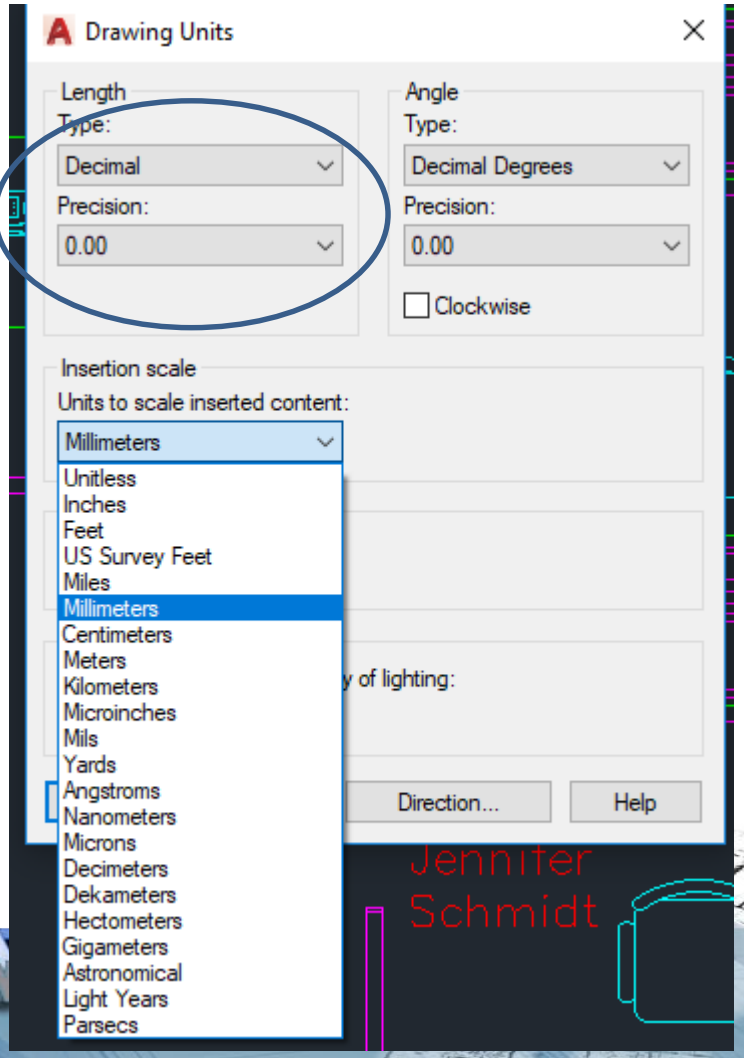


Drawing Utilities



Unit


Drawing Units
Length
Type:
Architectural
Precision:
0'-0 1/64"
Angle
Type:
Decimal Degrees
Precision:
0.00
Clockwise
Insertion scale
Units to scale inserted content:
Millimeters
Unitless
Inches
Feet
US Survey Feet
Miles
Millimeters
Centimeters
Meters
Kilometers
Microinches
Mils
Yards
Angstroms
Nanometers
Microns
Decimeters
Dekameters
Hectometers
Gigameters
Astronomical
Light Years
Parsecs
of lighting:
Direction...
Help

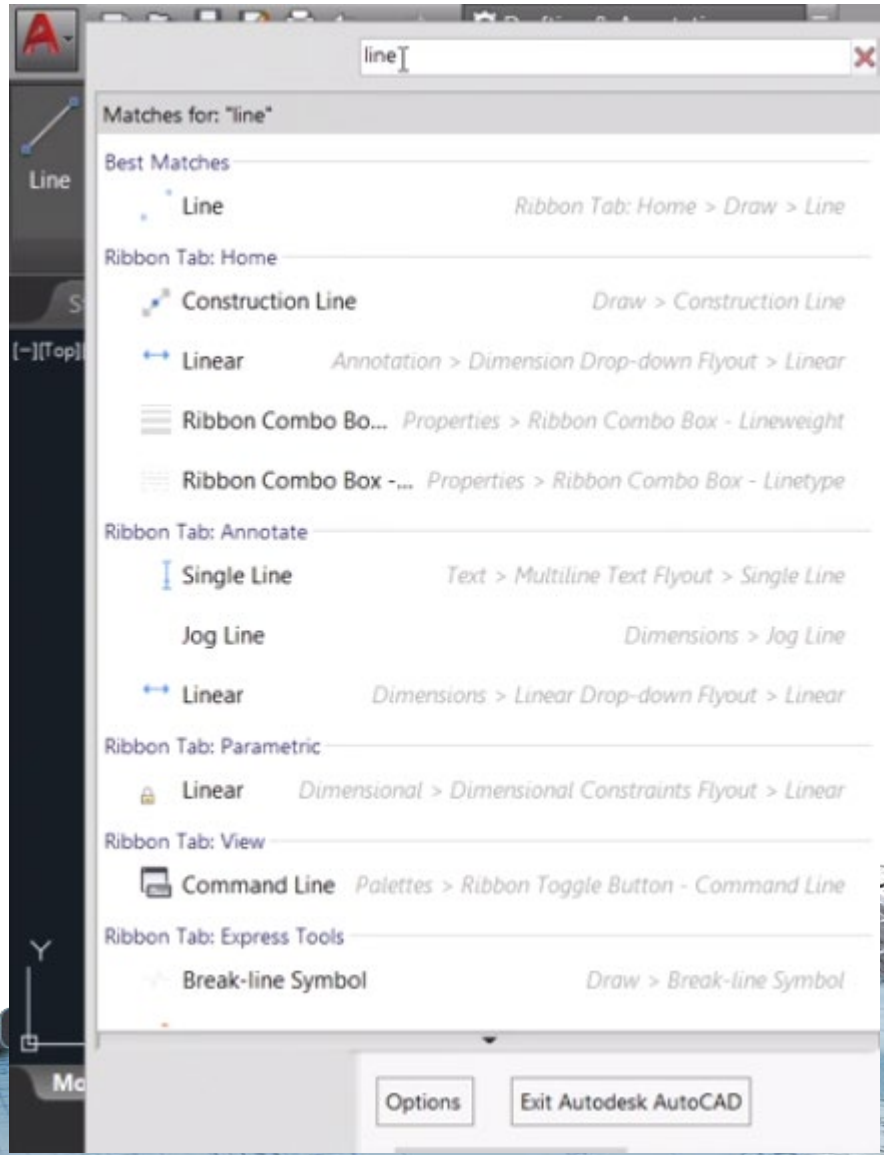

Drawing Units
Length
Type:
Decimal
Precision:
0.00
Angle
Type:
Decimal Degrees
Precision:
0.00
Clockwise
Insertion scale
Units to scale inserted content:
Millimeters
Unitless
Inches
Feet
US Survey Feet
Miles
Millimeters
Centimeters
Meters
Kilometers
Microinches
Mils
Yards
Angstroms
Nanometers
Microns
Decimeters
Dekameters
Hectometers
Gigameters
Astronomical
Light Years
Parsecs
of lighting:
Direction...
Help
Jenniter
Schmidt

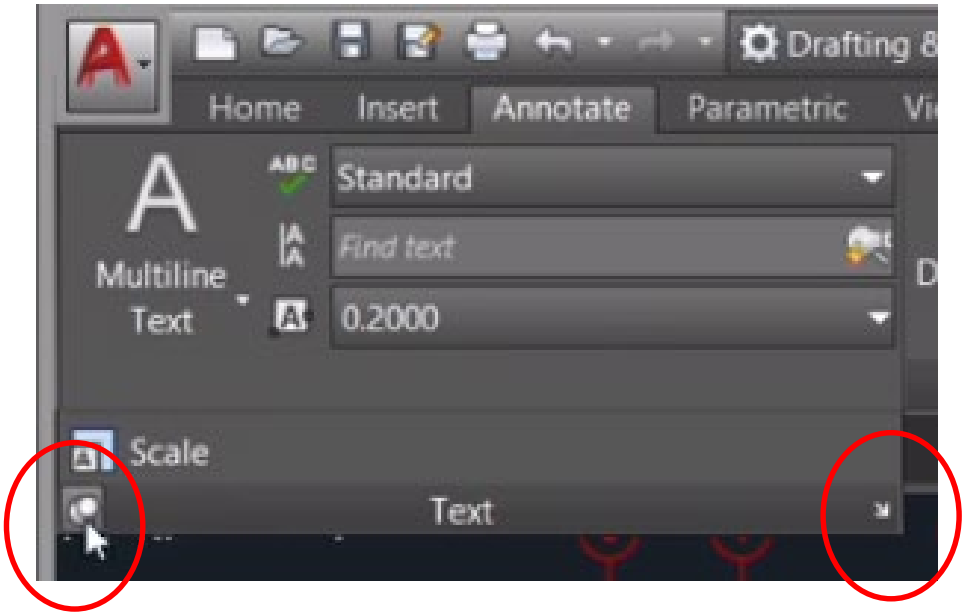
Basic tools

Click 

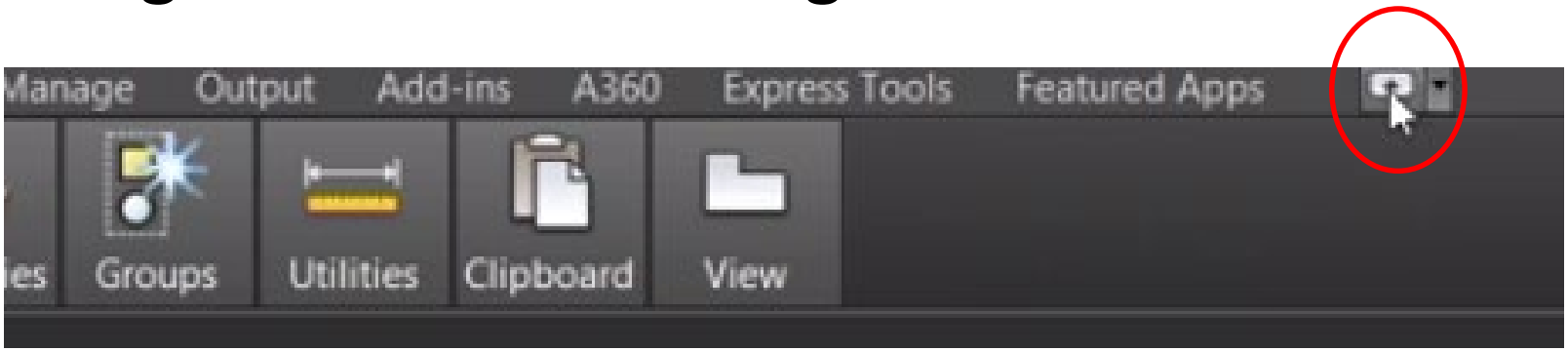


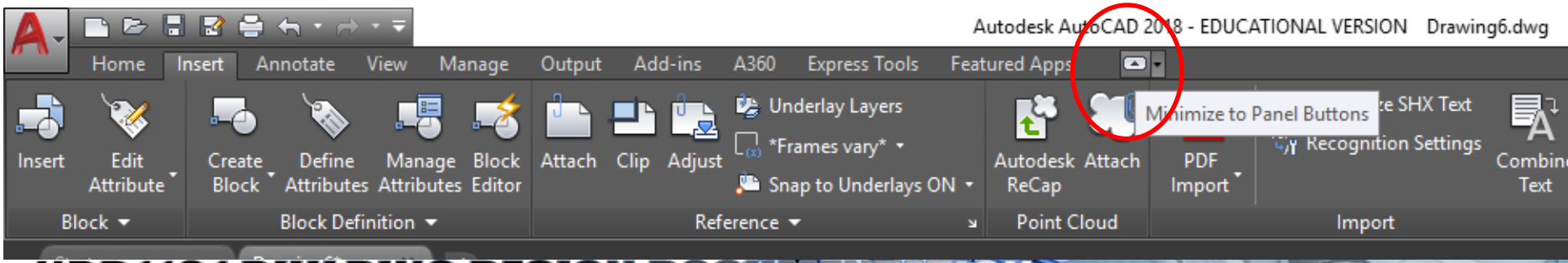
Type in ‘line’



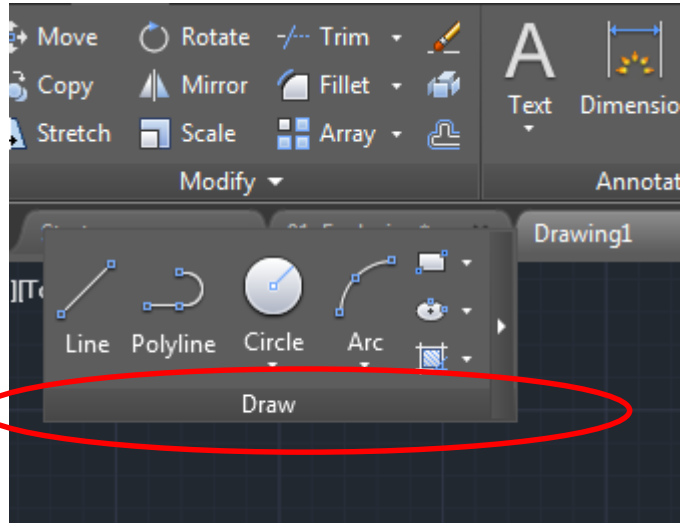

Drafting &
Home
Insert
Annotate
Parametric
Standard
Find text
Multiline Text
0.2000
Scale
Text

- Change the ribbon setting

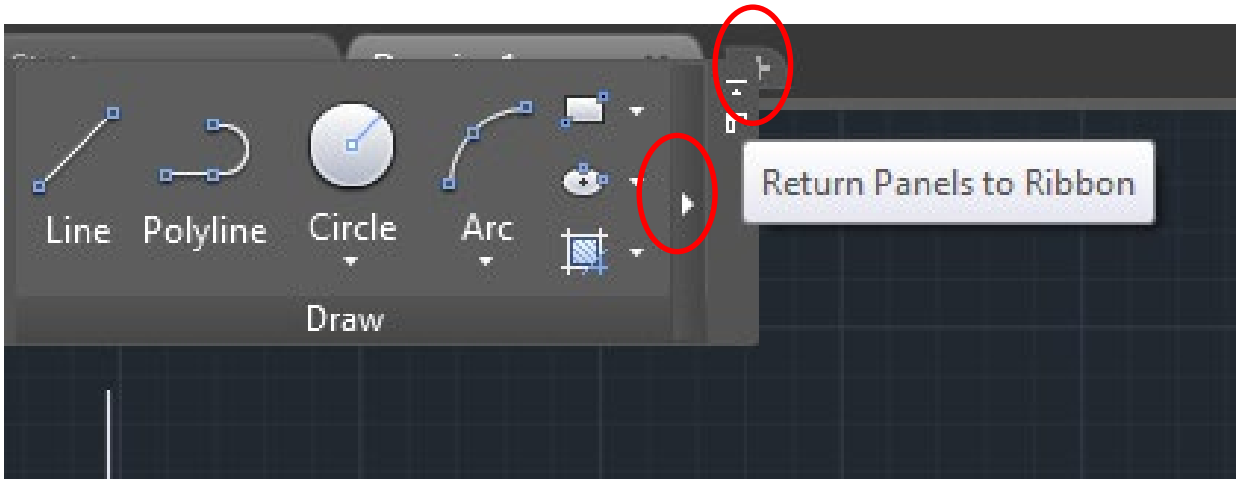




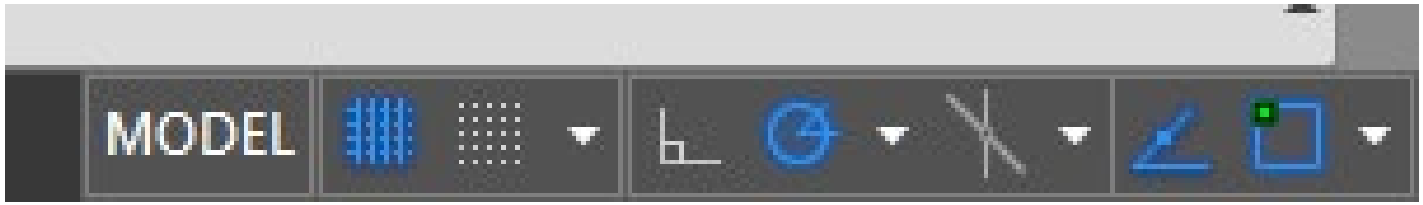
- Drag down the command panel as a floating window



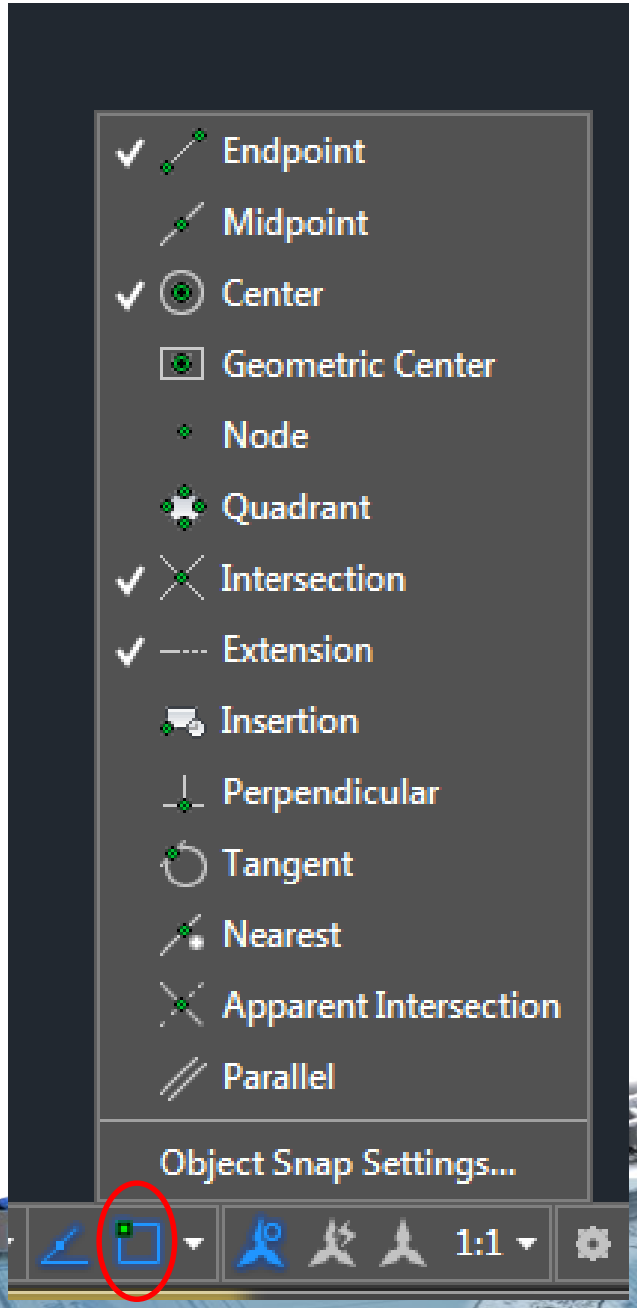
- Return panels to ribbon



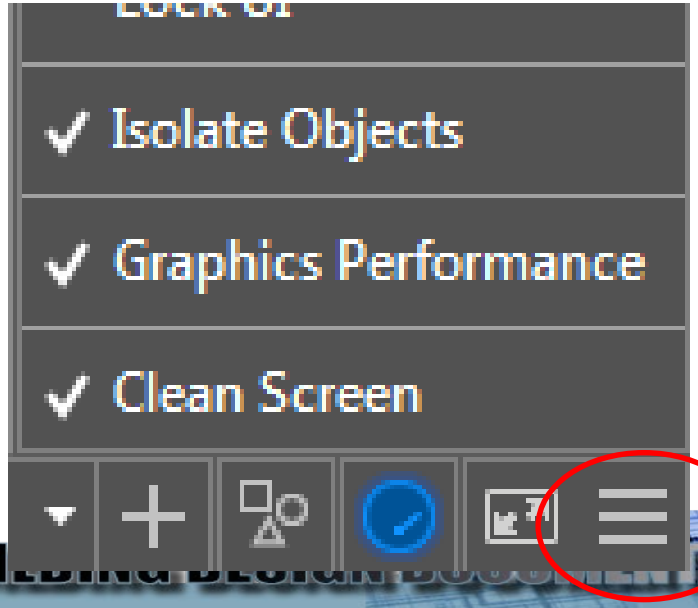
- Explore the status bar

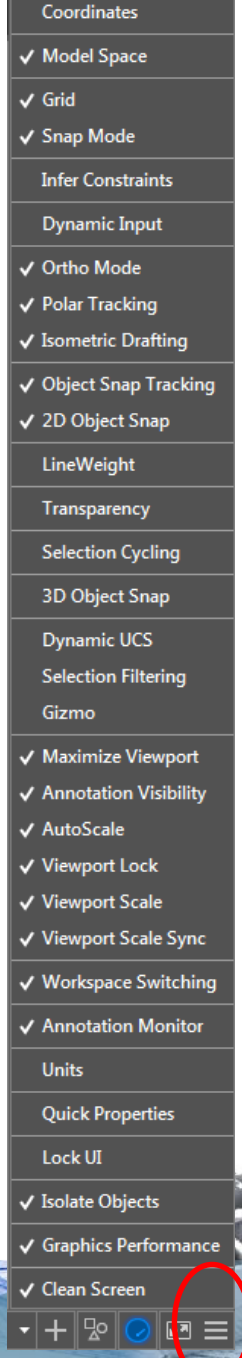


- Object Snap Setting
  - Right click



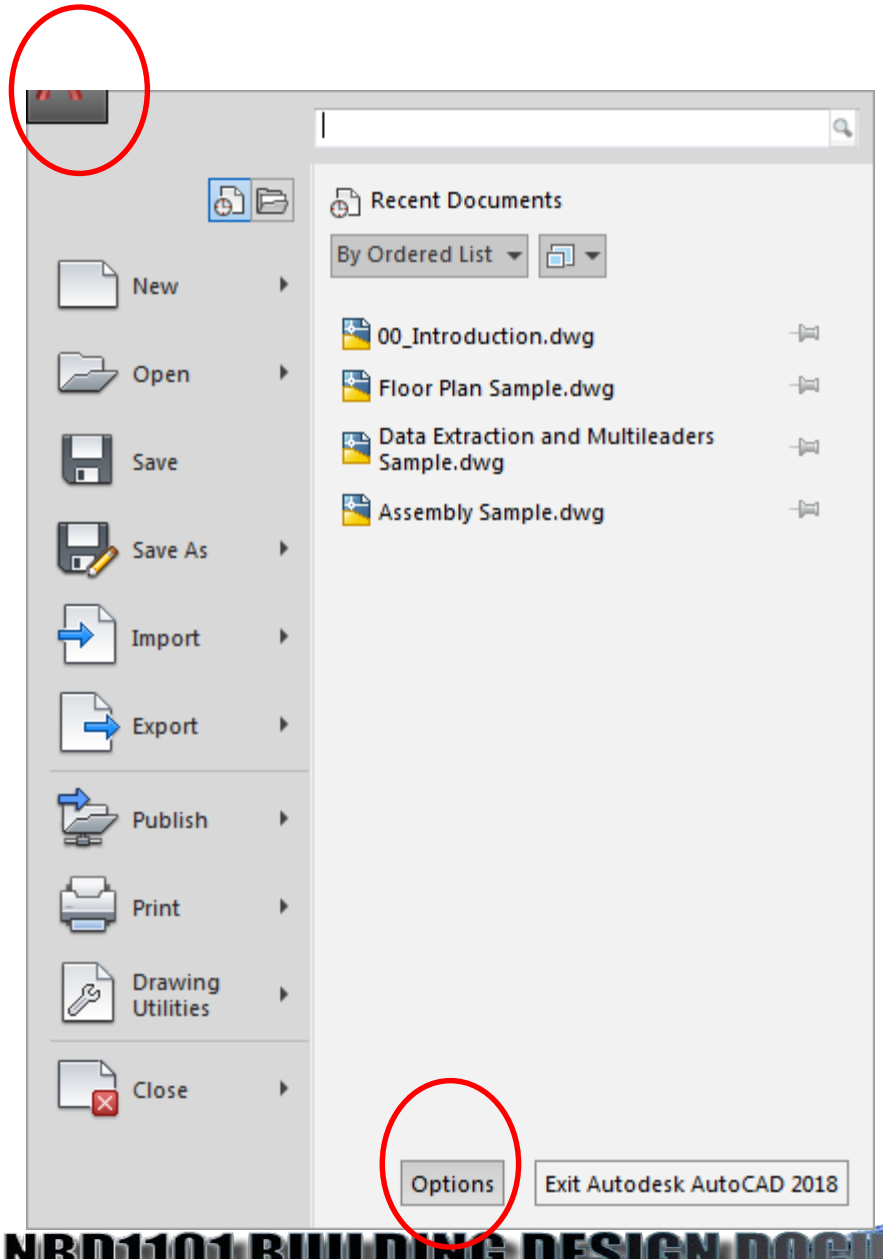
- Customise the status bar



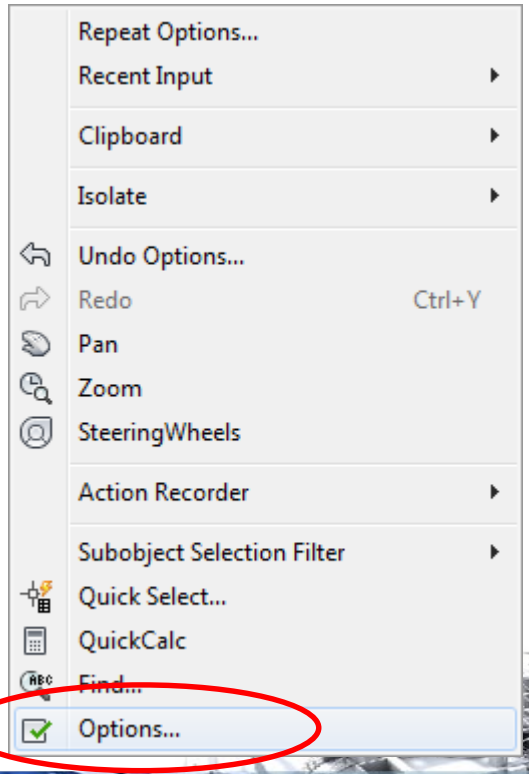


Set up workspace

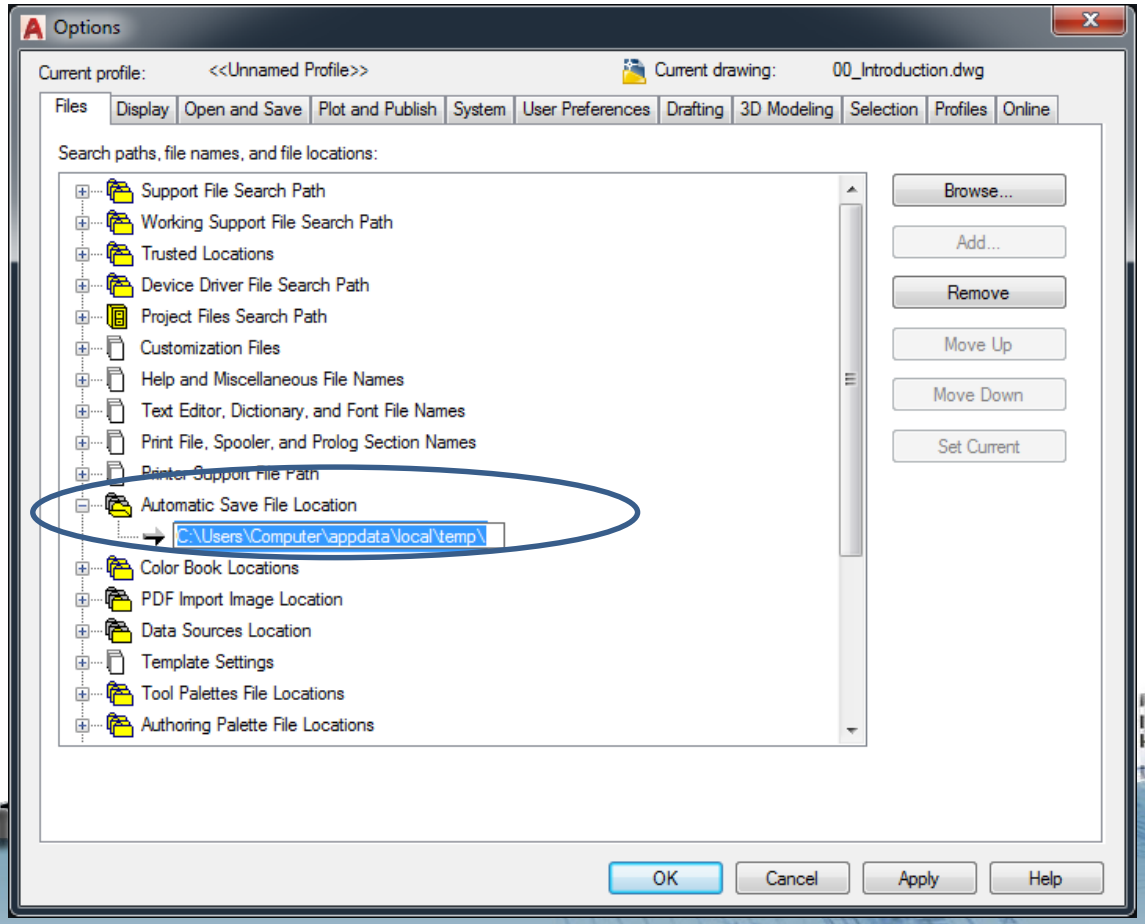




- OR right click the drawing area

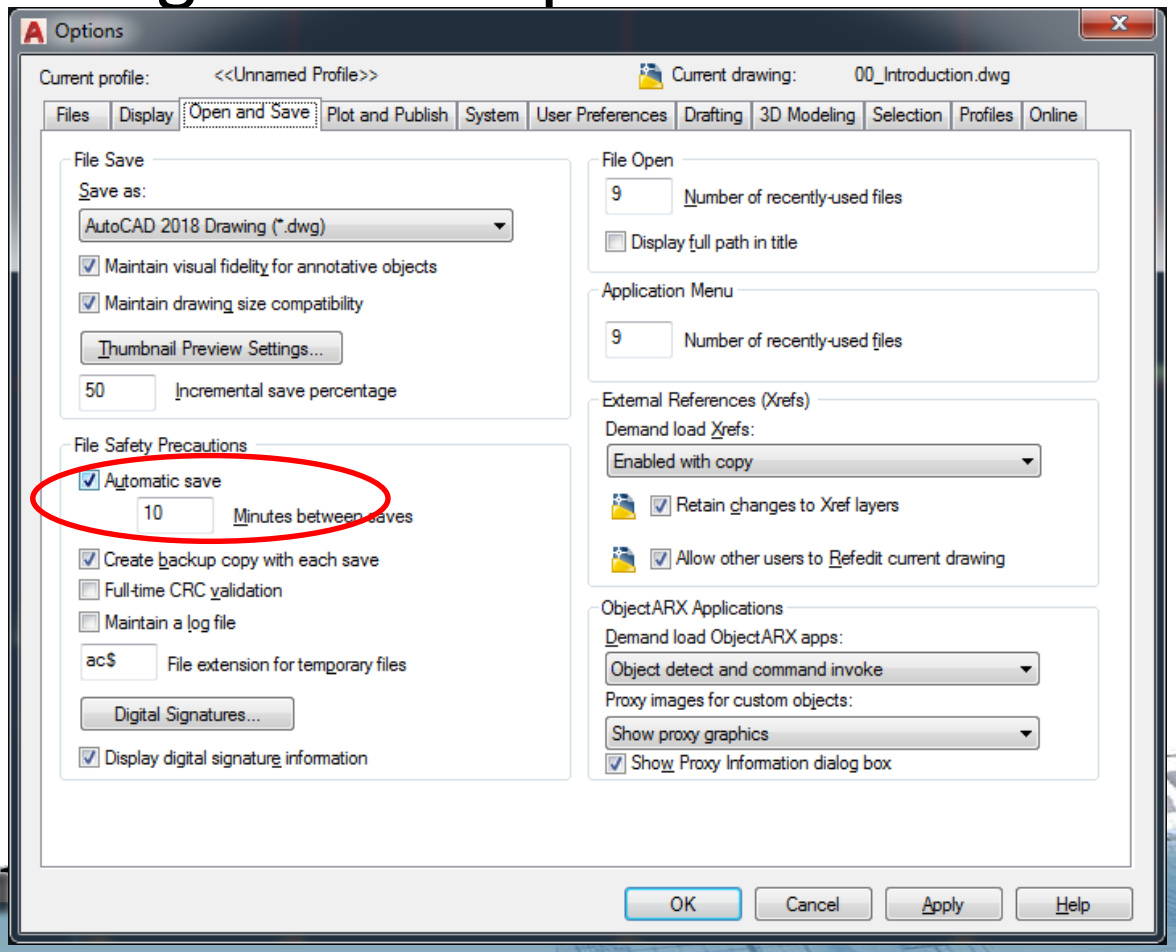


- Change the location of auto saving

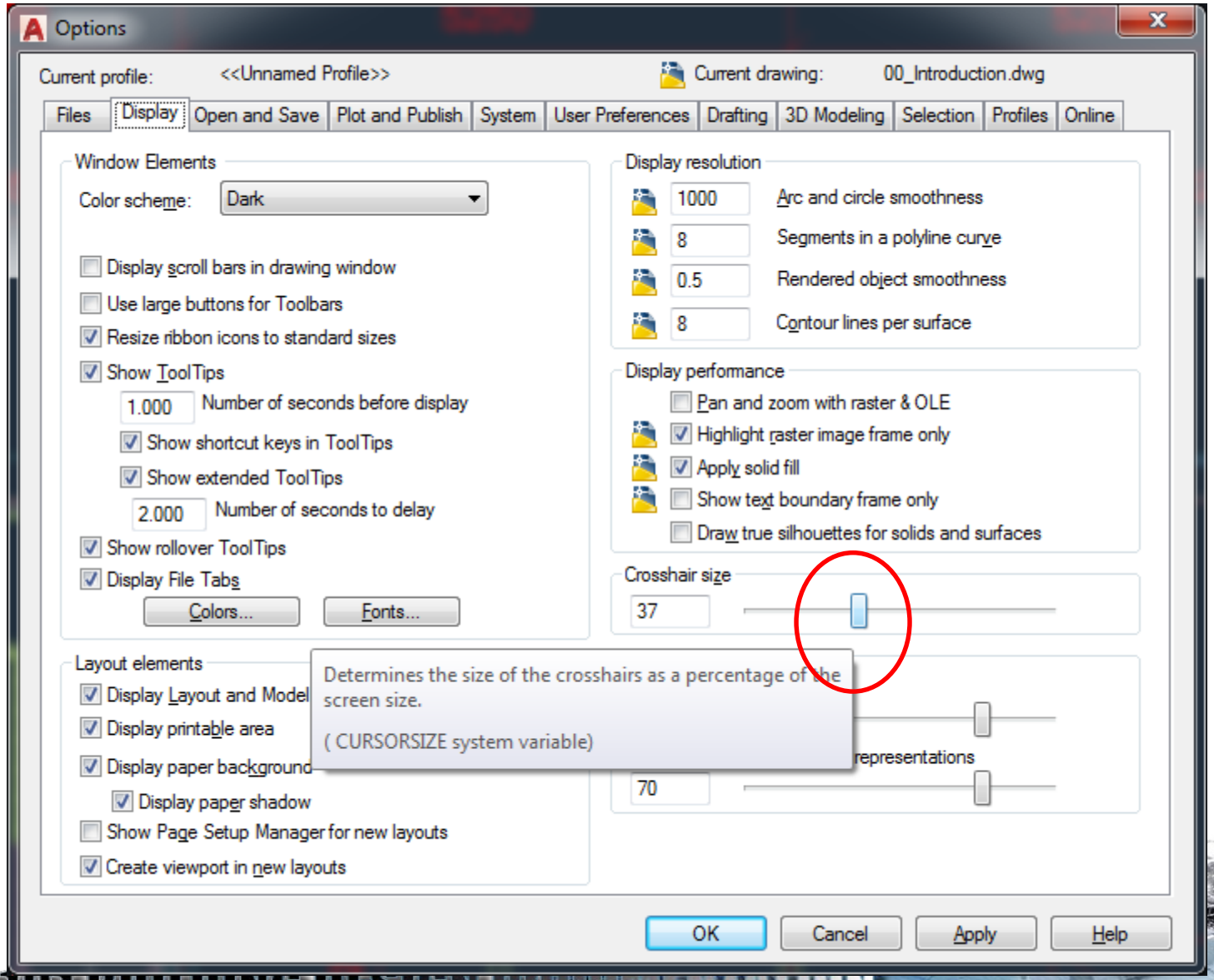


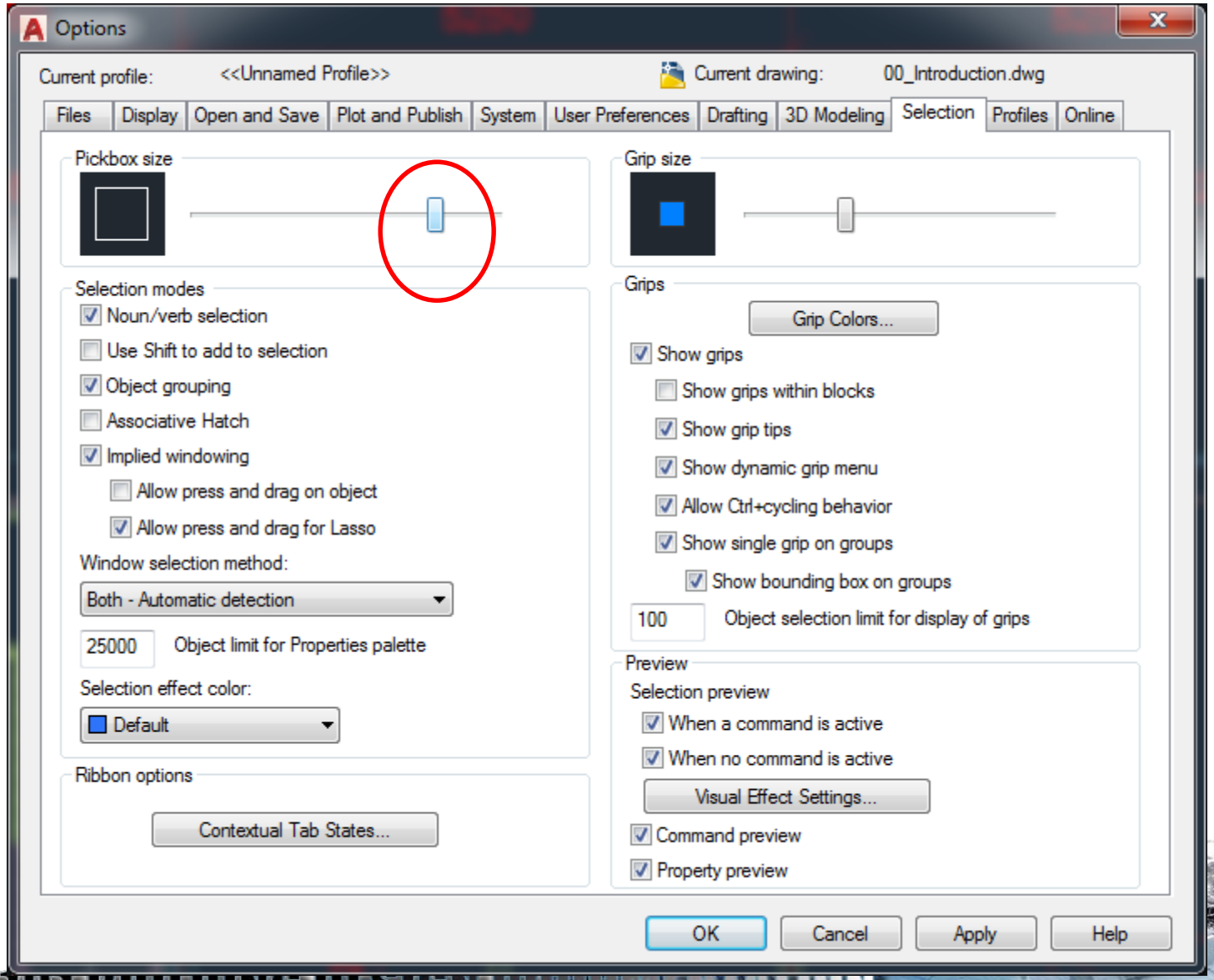


- Setting for backup file

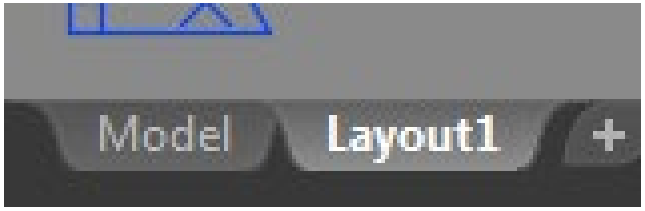


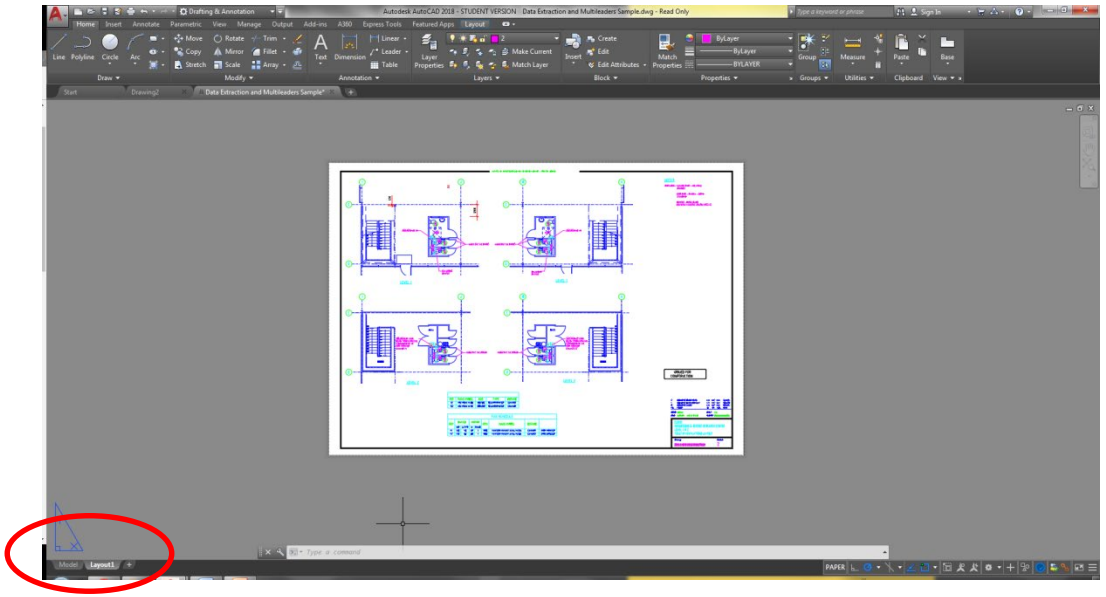



Options
Current profile: <<Unnamed Profile>>
Current drawing: 00_Introduction.dwg
Files
Display
Open and Save
Plot and Publish
System
User Preferences
Drafting
3D Modeling
Selection
Profiles
Online
Window Elements
Color scheme: Dark
Display scroll bars in drawing window
Use large buttons for Toolbars
Resize ribbon icons to standard sizes
Show ToolTips
1.000 Number of seconds before display
Show shortcut keys in ToolTips
Show extended ToolTips
2.000 Number of seconds to delay
Show rollover ToolTips
Display File Tabs
Colors...
Fonts...
Layout elements
Display Layout and Model
Display printable area
Display paper background
Display paper shadow
Show Page Setup Manager for new layouts
Create viewport in new layouts
Display resolution
1000 Arc and circle smoothness
8 Segments in a polyline curve
0.5 Rendered object smoothness
8 Contour lines per surface
Display performance
Pan and zoom with raster & OLE
Highlight raster image frame only
Apply solid fill
Show text boundary frame only
Draw true silhouettes for solids and surfaces
Crosshair size
37
Determines the size of the crosshairs as a percentage of the screen size.
(CURSORSIZE system variable)
representations
70
OK
Cancel
Apply
Help

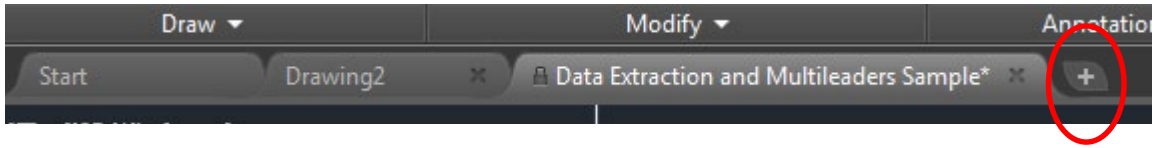

Options
Current profile: <<Unnamed Profile>>
Current drawing: 00_Introduction.dwg
Files
Display
Open and Save
Plot and Publish
System
User Preferences
Drafting
3D Modeling
Selection
Profiles
Online
Pickbox size
Grip size
Selection modes
Noun/verb selection
Use Shift to add to selection
Object grouping
Associative Hatch
Implied windowing
Allow press and drag on object
Allow press and drag for Lasso
Window selection method:
Both - Automatic detection
25000
Object limit for Properties palette
Selection effect color:
Default
Ribbon options
Contextual Tab States...
Grips
Grip Colors...
Show grips
Show grips within blocks
Show grip tips
Show dynamic grip menu
Allow Ctrl+cycling behavior
Show single grip on groups
Show bounding box on groups
100
Object selection limit for display of grips
Preview
Selection preview
When a command is active
When no command is active
Visual Effect Settings...
Command preview
Property preview
OK
Cancel
Apply
Help

- View the layout of the drawing





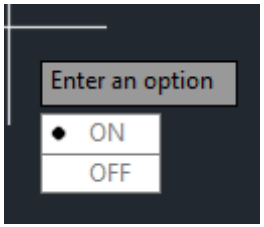
- Create a new drawing project

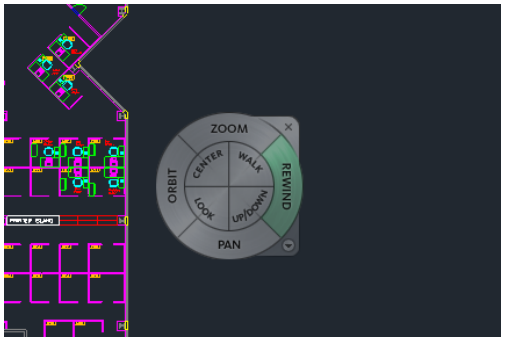


- NAVBAR

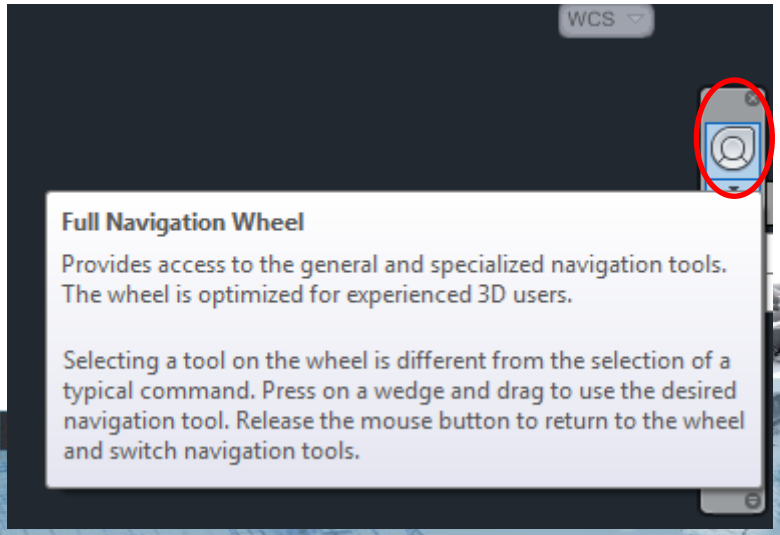


- Type in ‘navbar’





- Click the Full Navigation Wheel and move it out

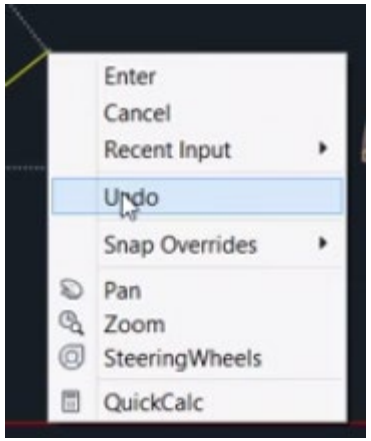


- Play with the view cube



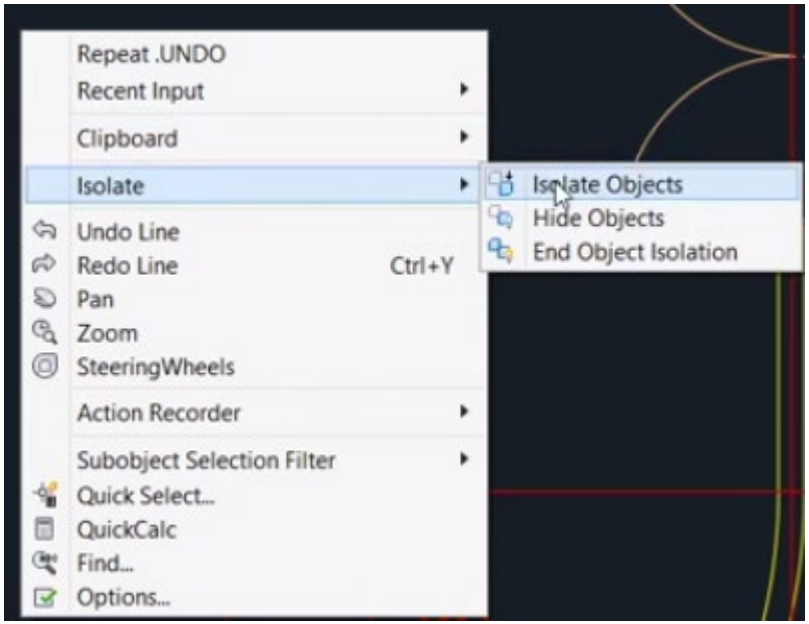
Shortcut menu

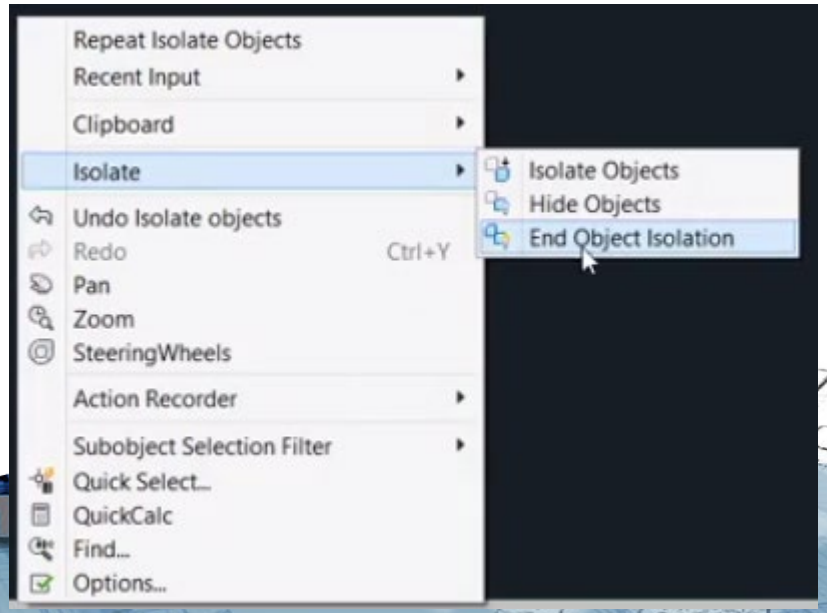
- Shortcut menu
  - Eg. Draw under the ‘line’ command
  - Right click the mouse



Shortcut menu

- Right click

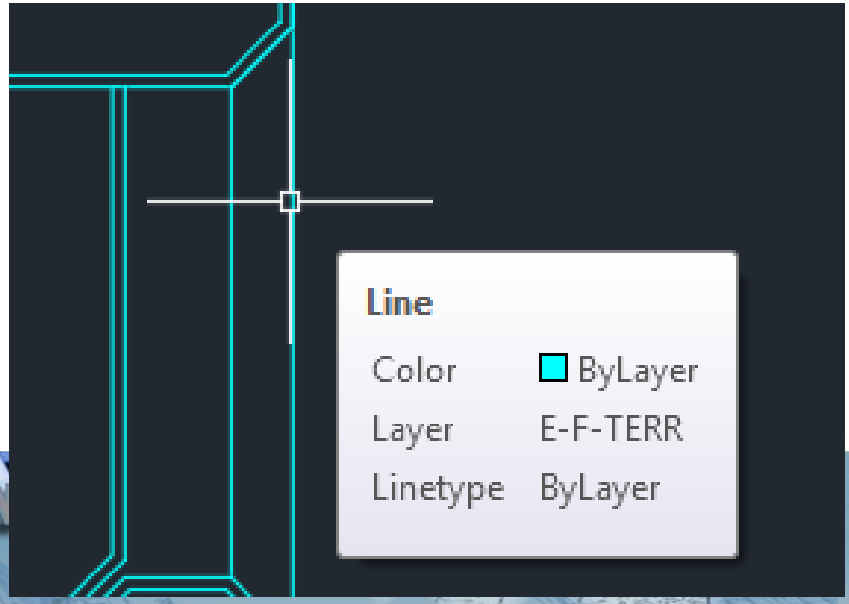




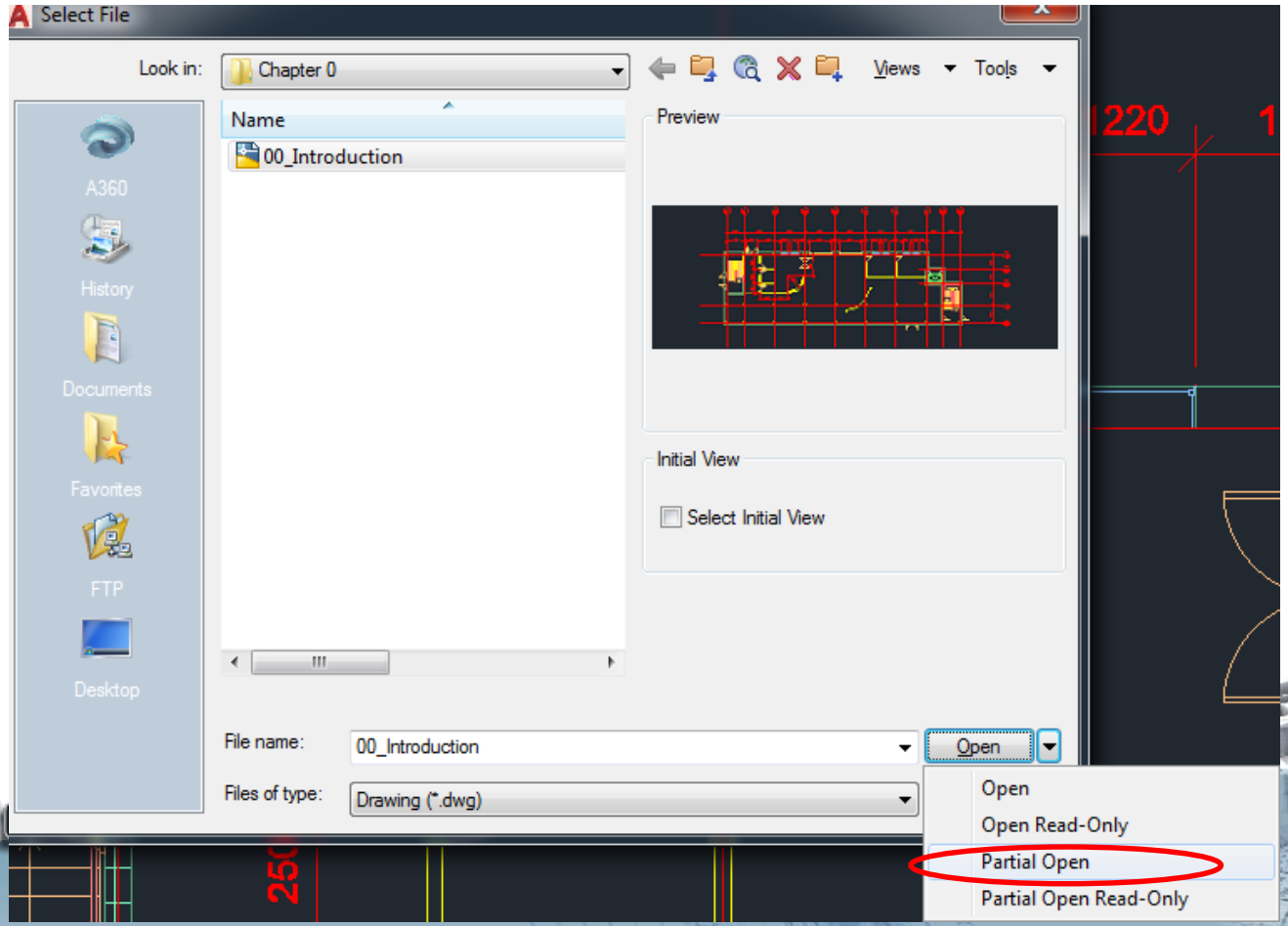
- Play with different shortcut menu
- eg. In ‘Polyline”



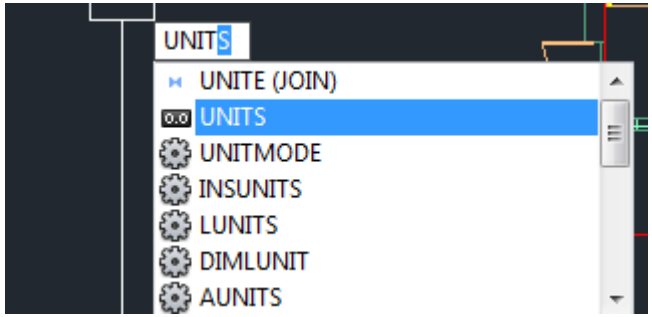
- Object information
  - Hove the mouse upon an object
  - Layer
  - Try to change an object's layer

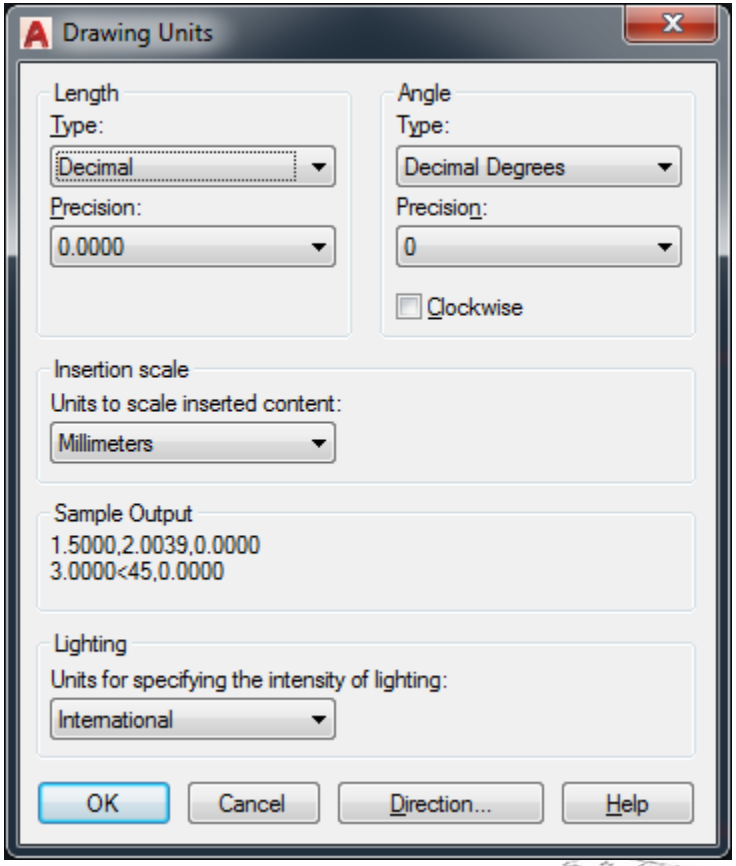


- Select certain layers to be open

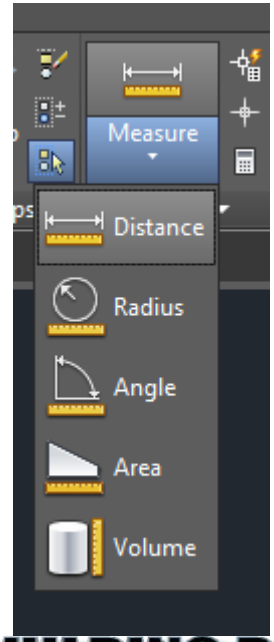


- Type ‘units’

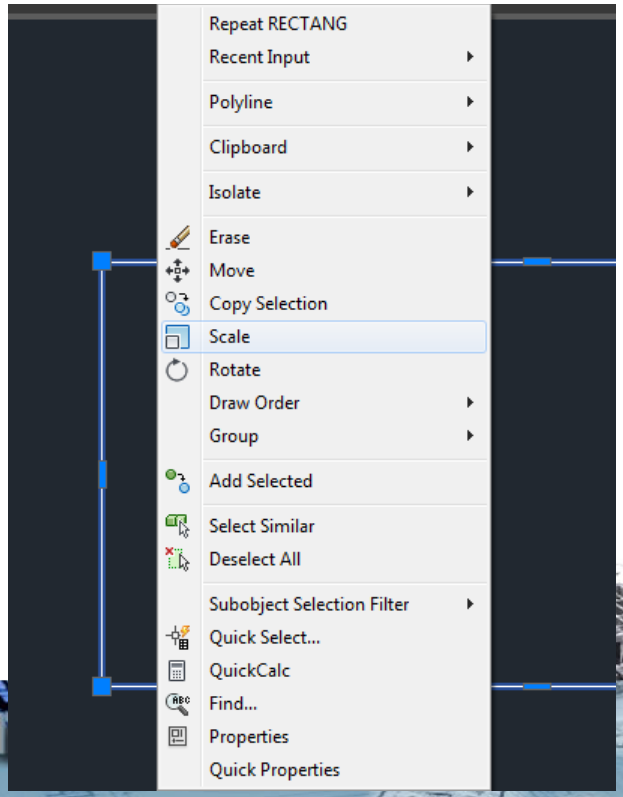

UNITS
UNITE (JOIN)
UNITS
UNITMODE
INSUNITS
LUNITS
DIMLUNIT
AUNITS


Drawing Units
Length
Type:
Decimal
Precision:
0.0000
Angle
Type:
Decimal Degrees
Precision:
0
Clockwise
Insertion scale
Units to scale inserted content:
Millimeters
Sample Output
1.5000,2.0039,0.0000
3.0000<45,0.0000
Lighting
Units for specifying the intensity of lighting:
International
OK
Cancel
Direction...
Help

- Measure



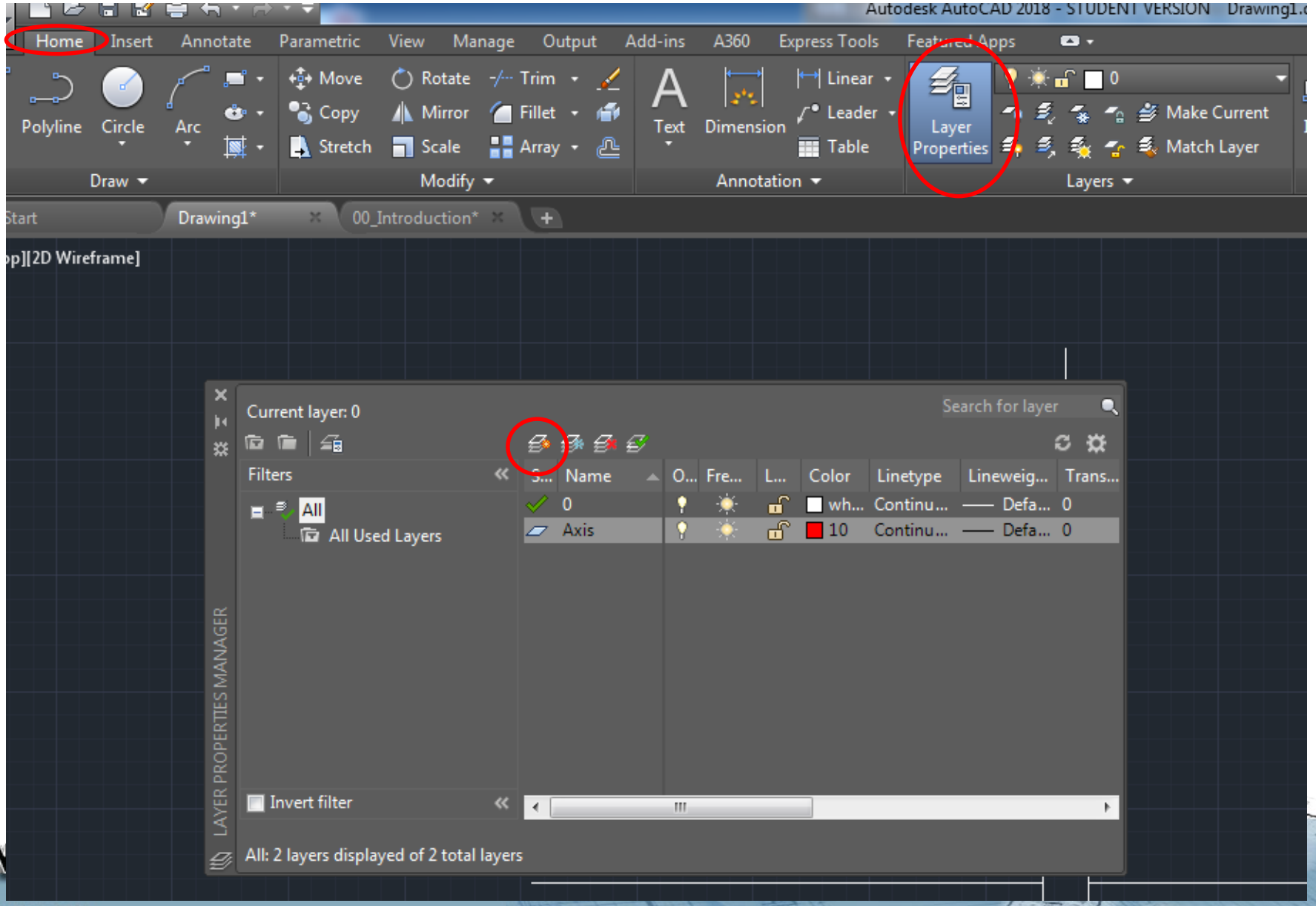
- Scale
  - Select an object; then right click
  - OR type ‘scale’



- .DWT: template
- ‘save as’-----save as a template
- (blank project)

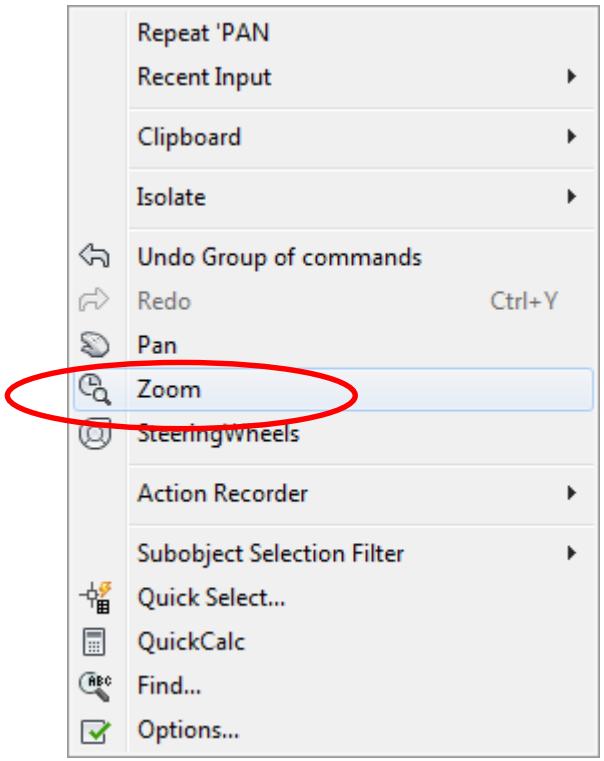


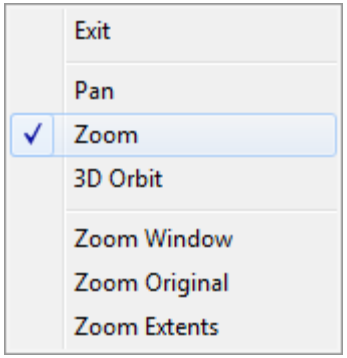
- Setting layers

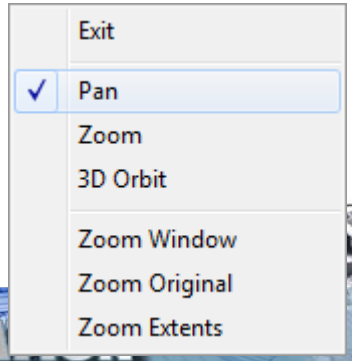


Navigating drawings

- Right click the drawing area



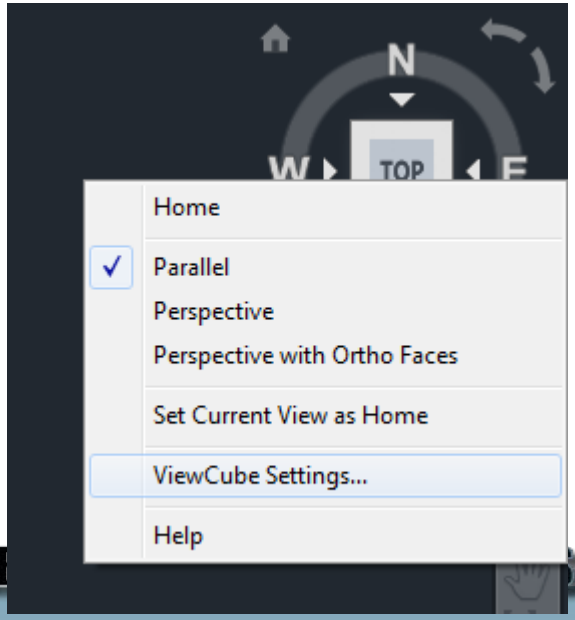


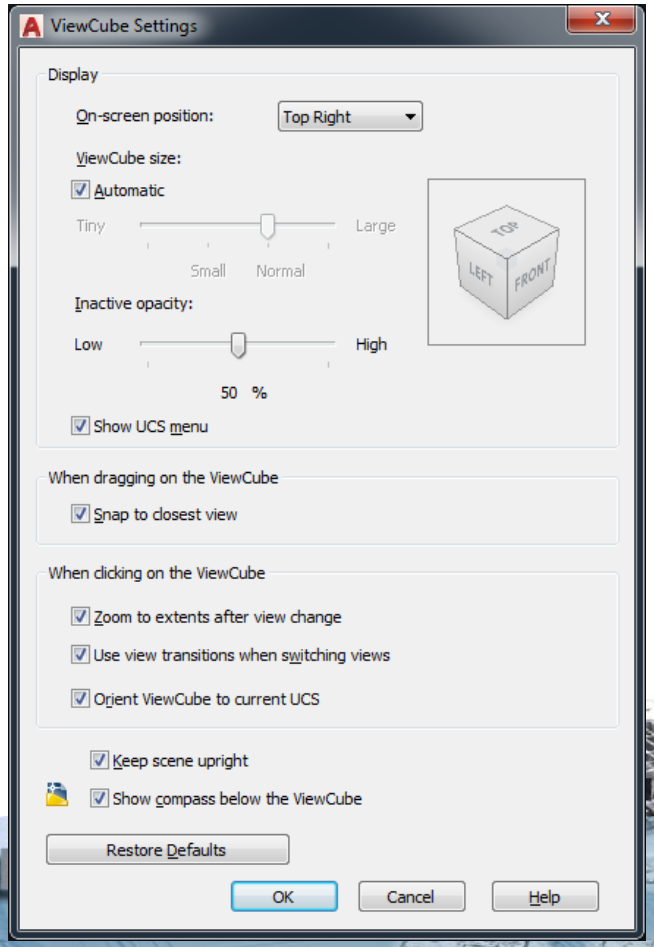


- The use of the mouse
- The use of mouse wheel
- Orbit: Shift key + Mouse wheel

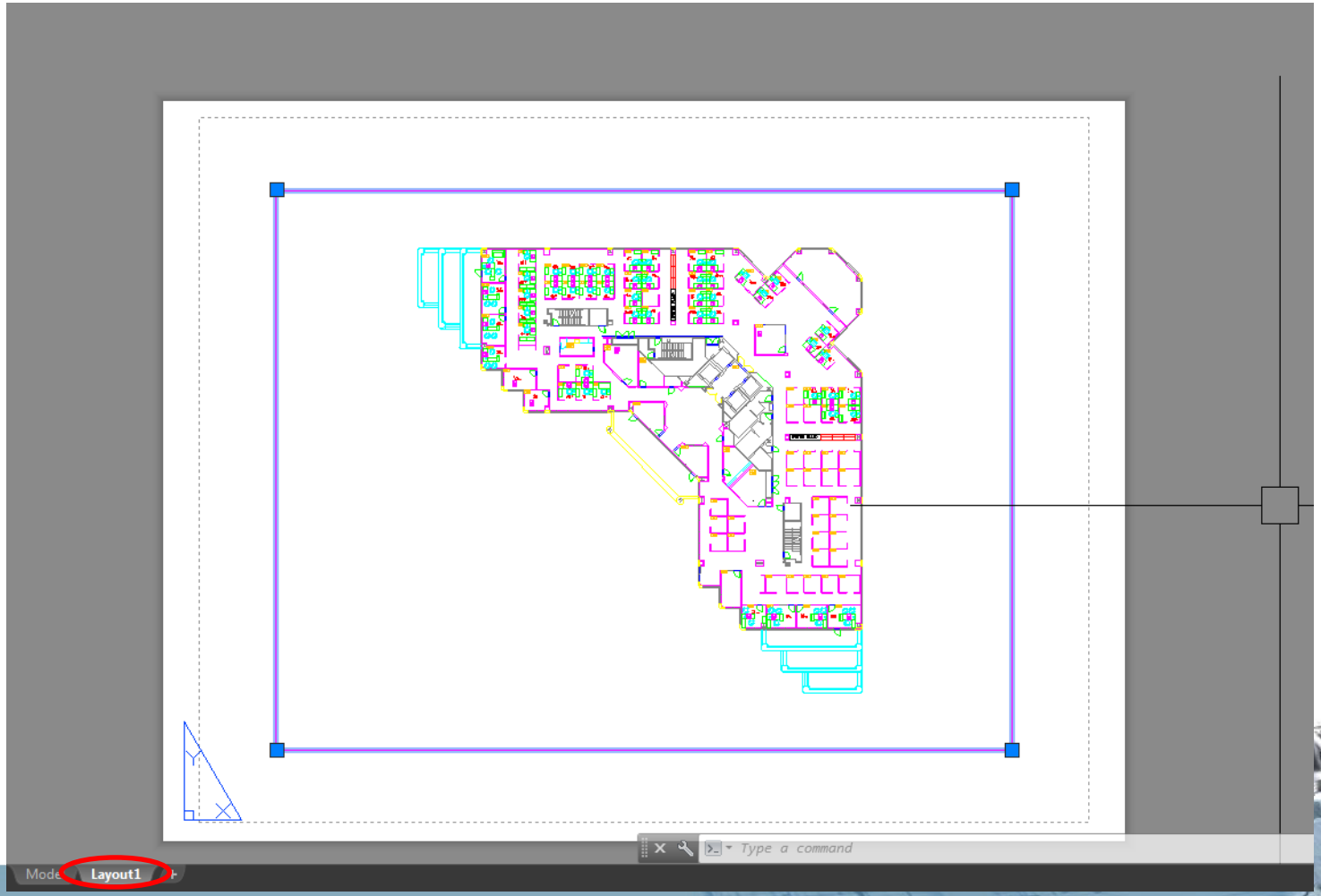


- ViewCube settings
  - Right click the ViewCube





Layout

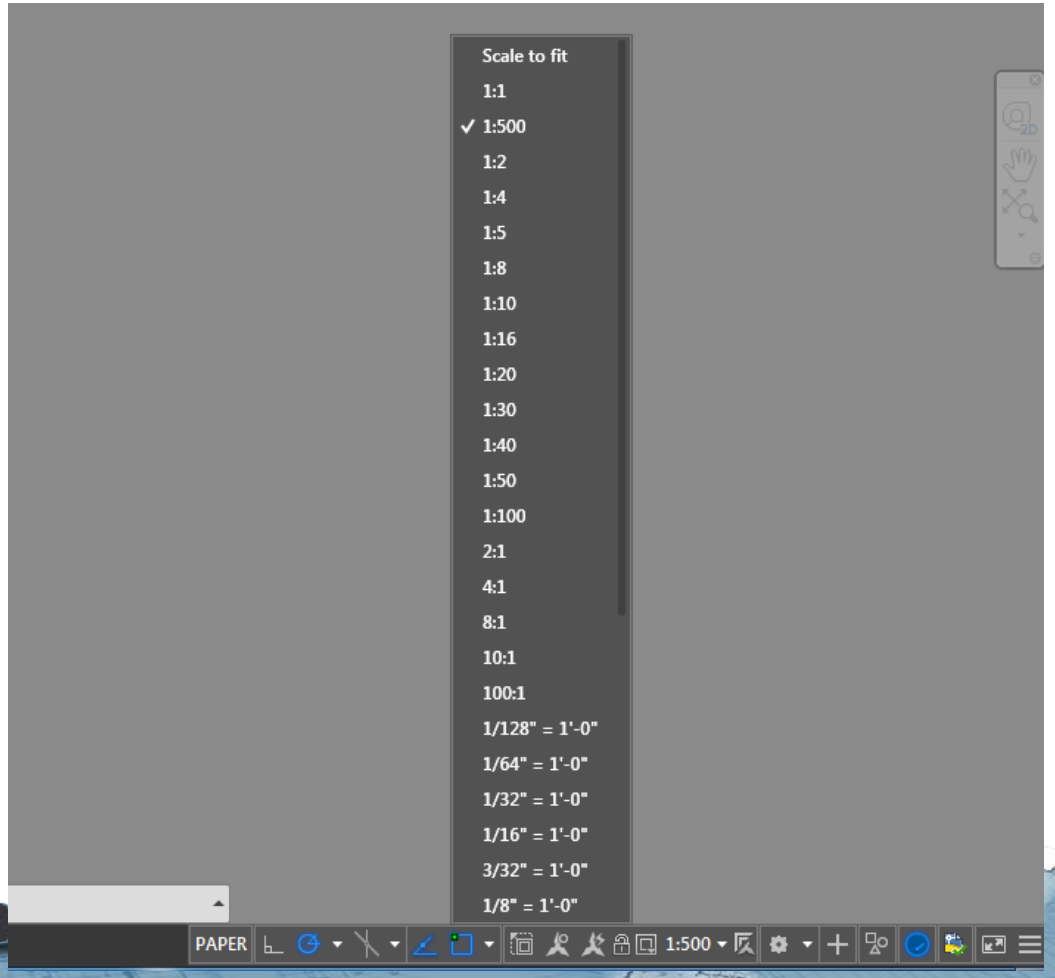


Go into ‘Layout’ view

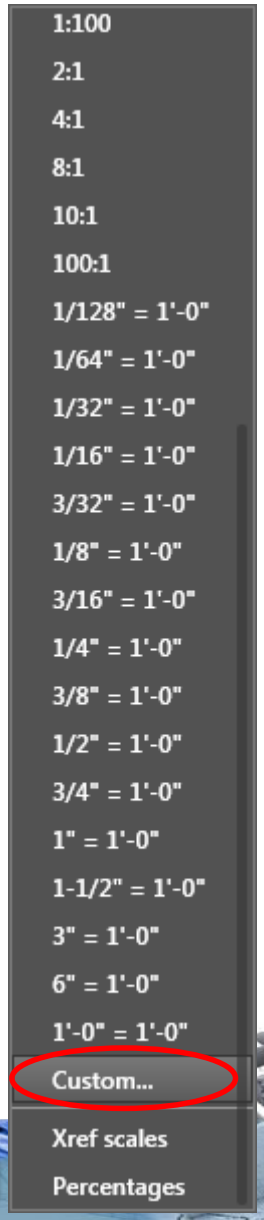
Click the frame to select the drawing

Layout

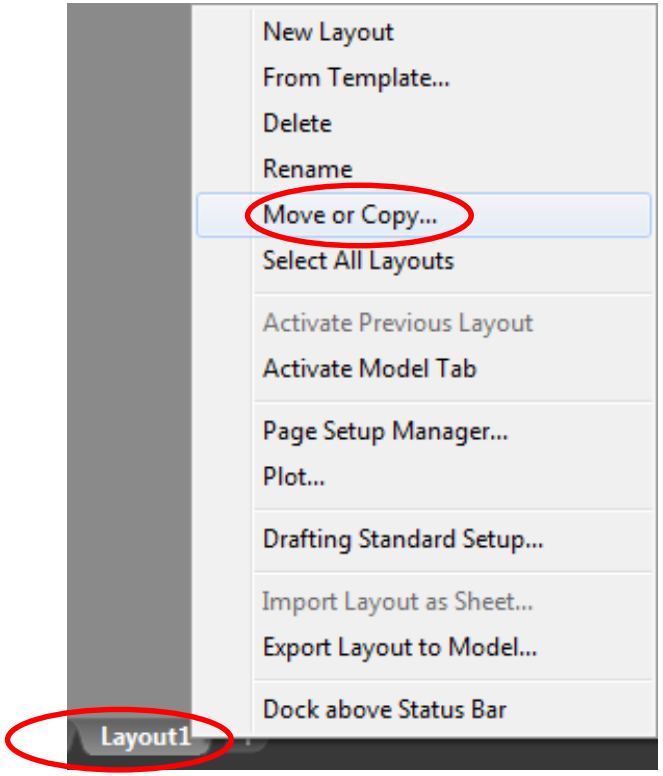
- Select a proper scale

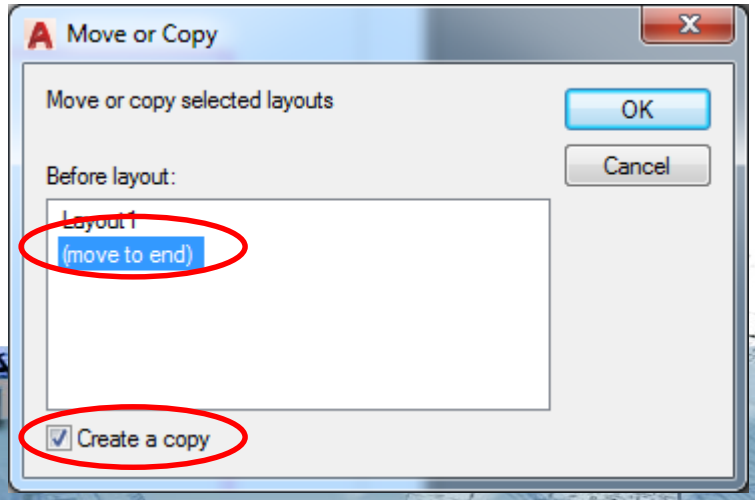


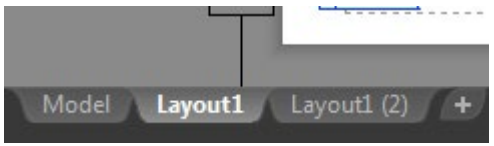
- If there is no proper scales
  - Customise a scale



- Copy the layout:
  - Right click the Layout, eg. Layout 1
  - Select ‘Move or Copy’
  - Select ‘(move to end)’
  - Check ‘Create a copy’

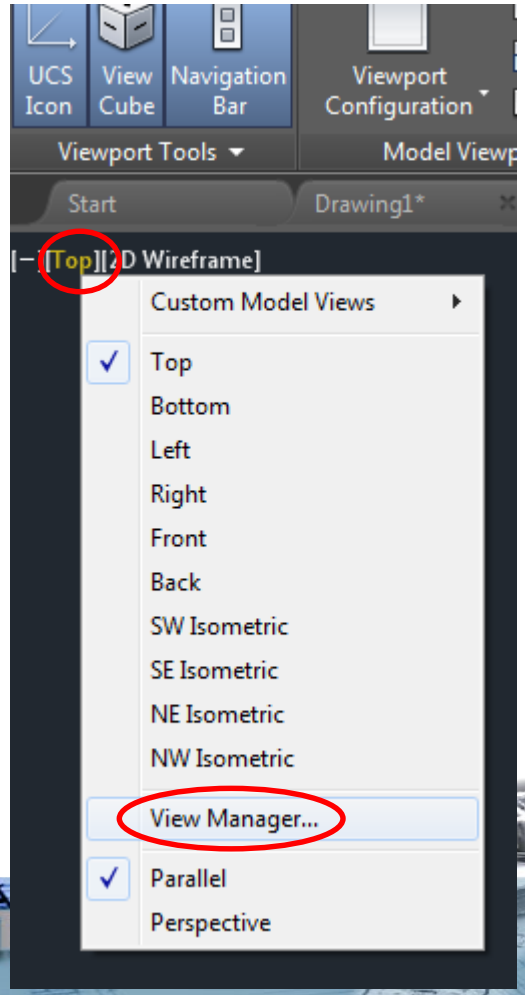






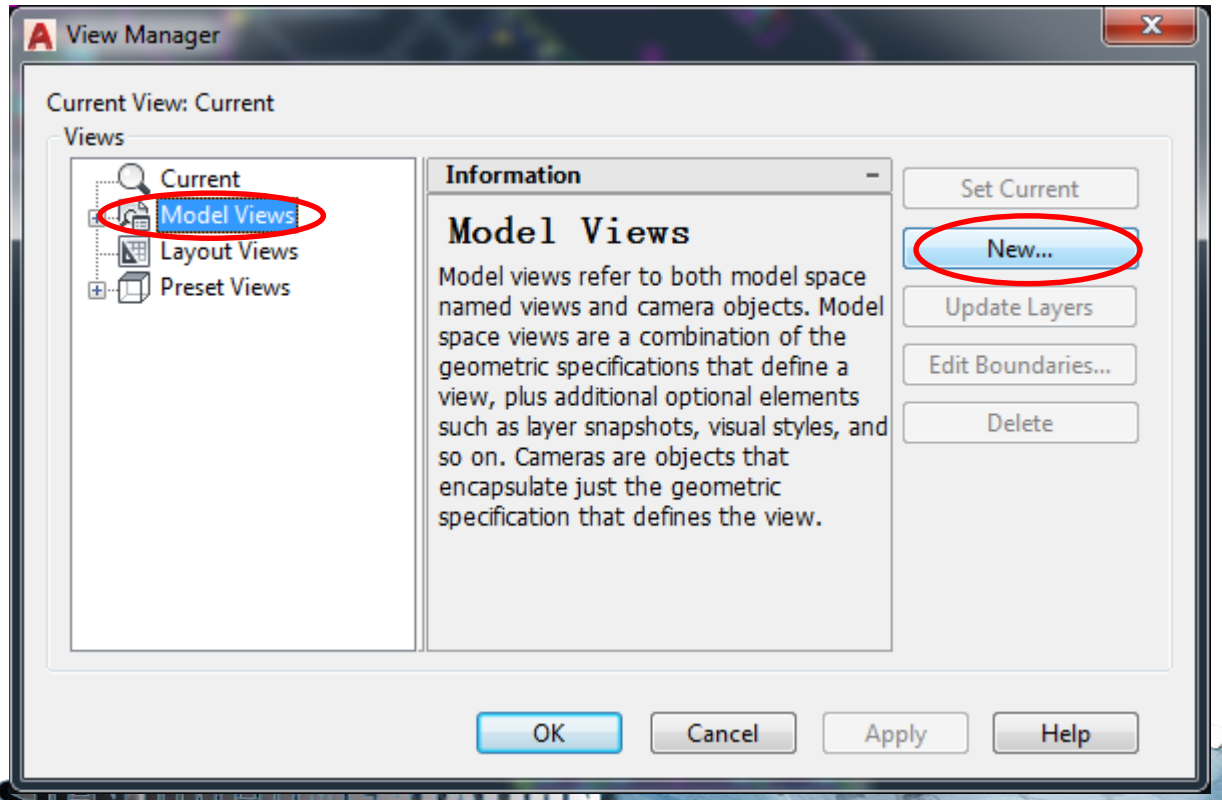
View manager

- Click ‘Top’
- Select ‘View Manager…’



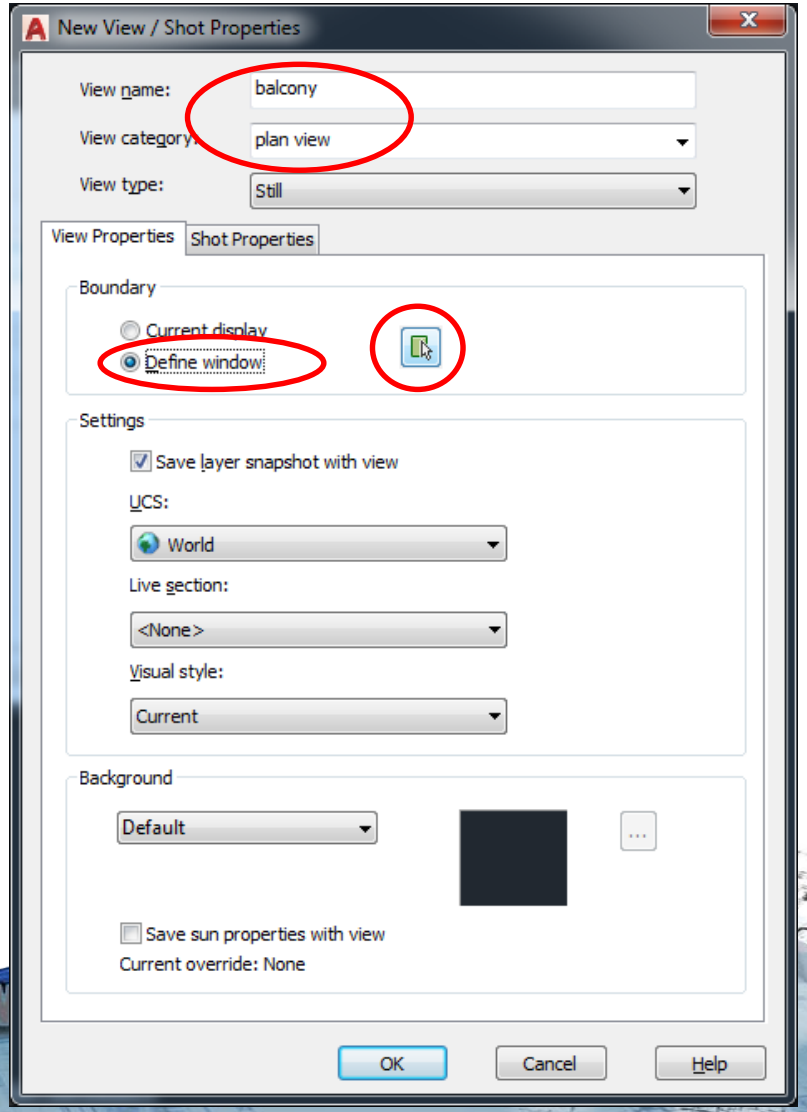
View manager

- Select ‘Model Views’
- Click ‘New…’



View manager

- Type in information
- Define window to select the view area

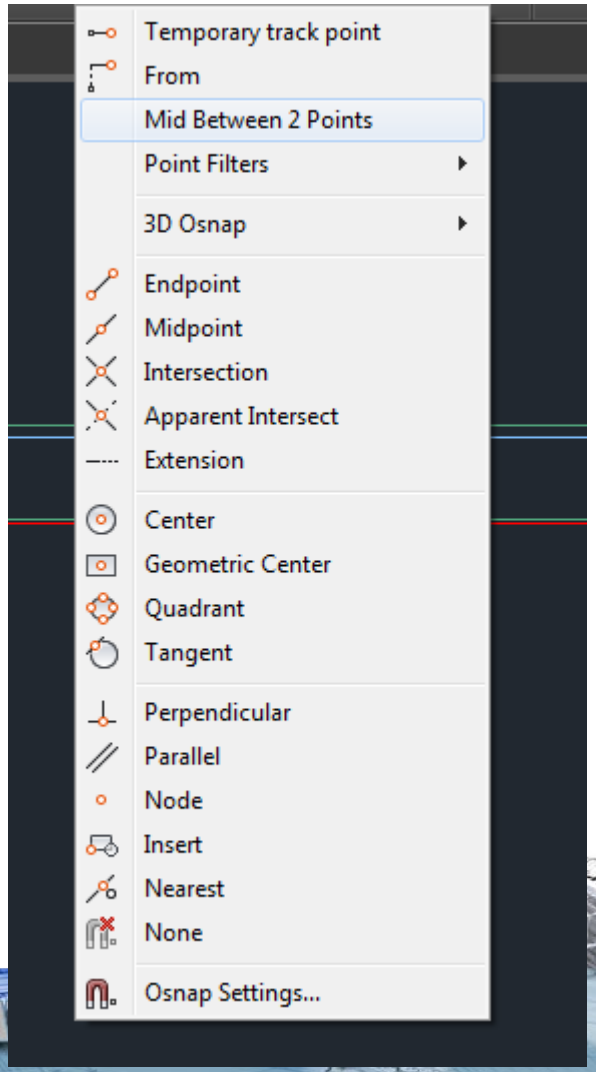


drawing



Drawing lines

- ‘Line’
- Shift + right click
- Eg. Select ‘Mid Between 2 Points’



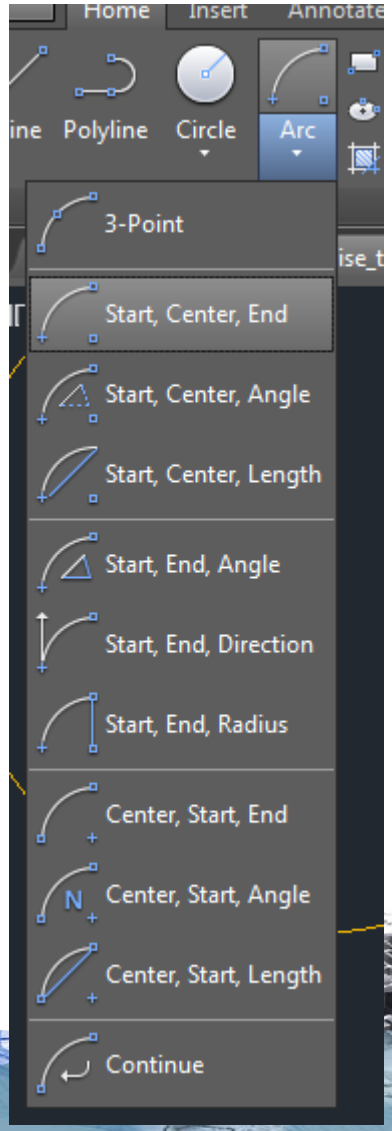
Drawing a circle

- Create a new layer, eg. Furniture
- Pay attention to the step of drawing a circle and the different options



Drawing arcs

- Layer, eg. I-wall
- Arc, eg. ‘Start, Center, End’
- Start point, diameter
- Change direction: hold Ctrl
- Type in a degree: can be negative, eg. -90



The use of drawing tools

- Ellipses
- Polylines
- Rectangles and polygons
- Points and donuts

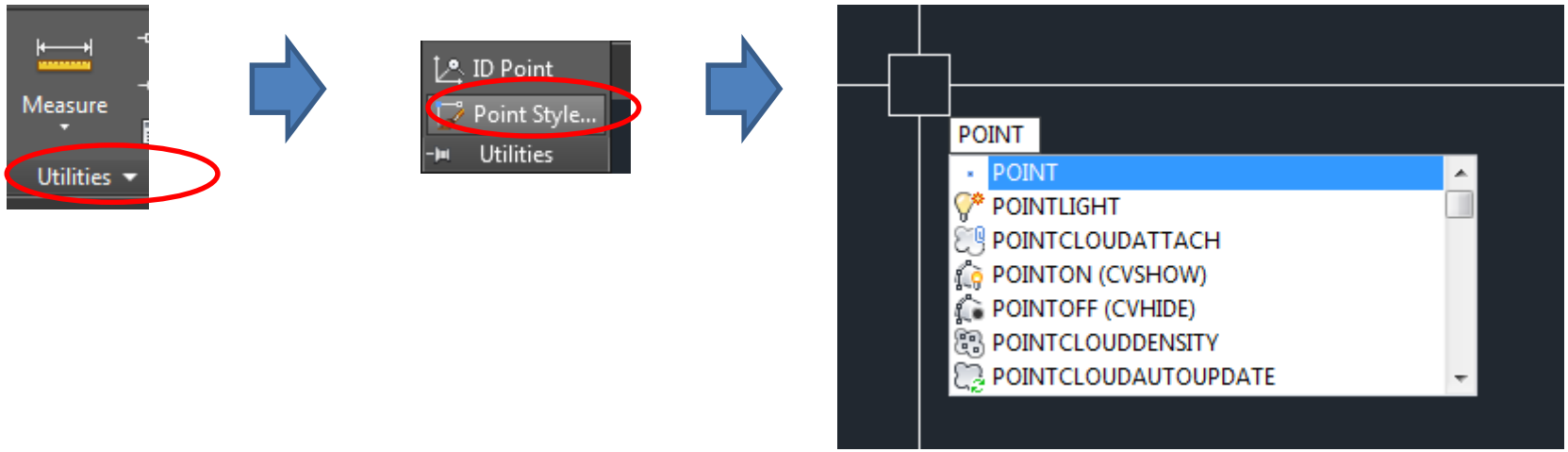


- Draw wall lines: ML (ml—scale define the width)



Drawing and editing points

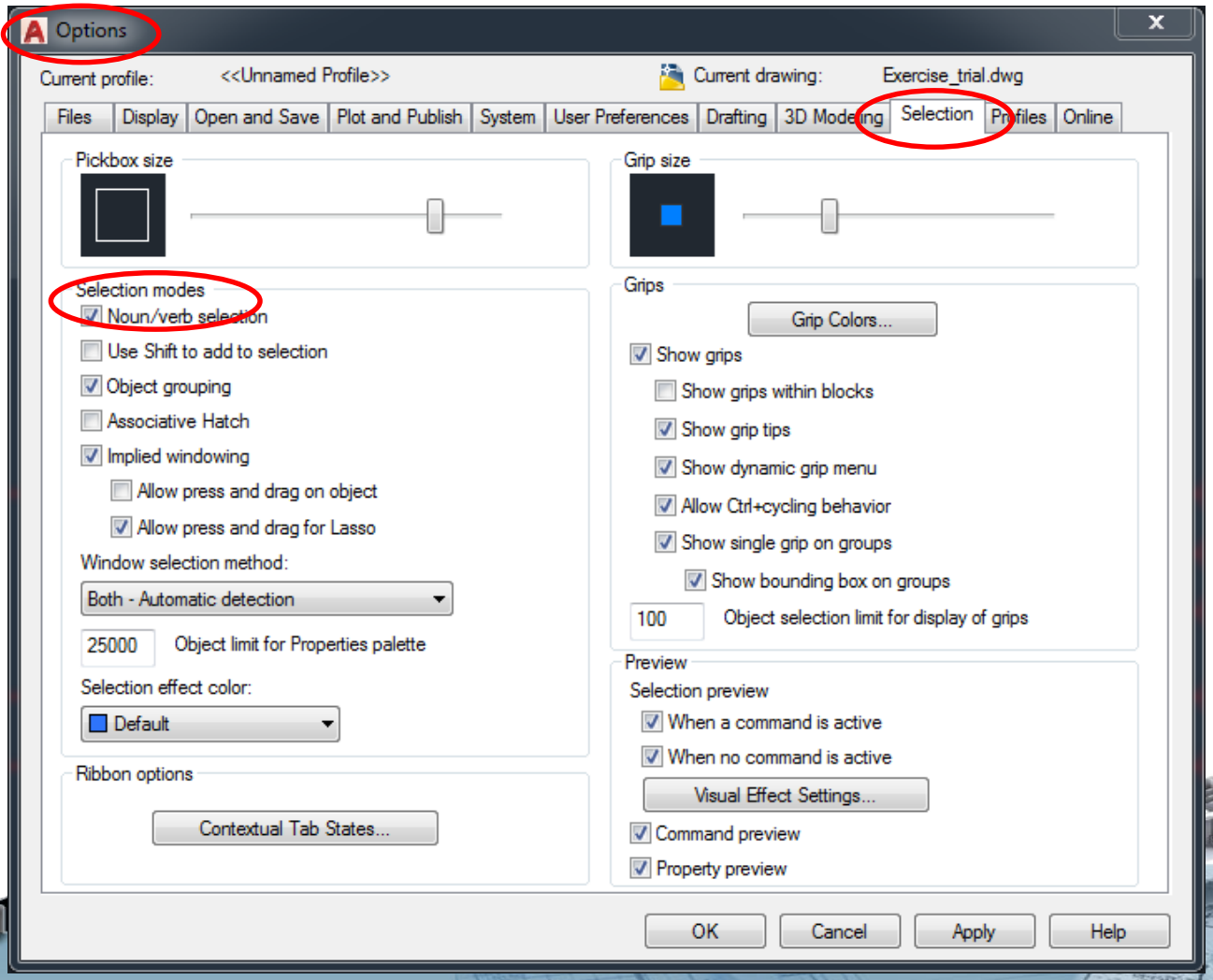
Type in ‘Point’



Editing and modifying



Selection setting



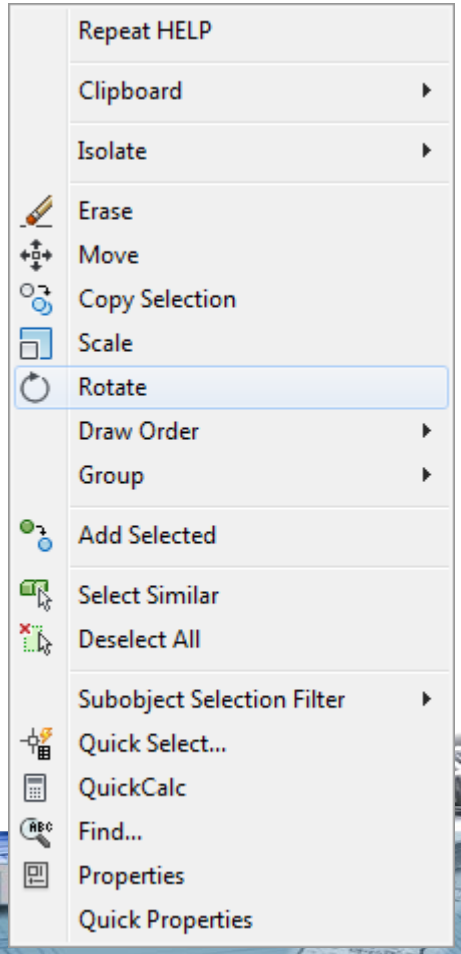
Different ways of selection

- Window selection
- Cross section

- Hold vs. release
- Towards right down vs. left up



- Select an object
- Right click
- A menu of editing actions available



Practise with

- Move and copy
- Scale and rotate
- Arrays
- Offset and mirror
- Trim and extend
- Stretch and lengthen
- Break and join



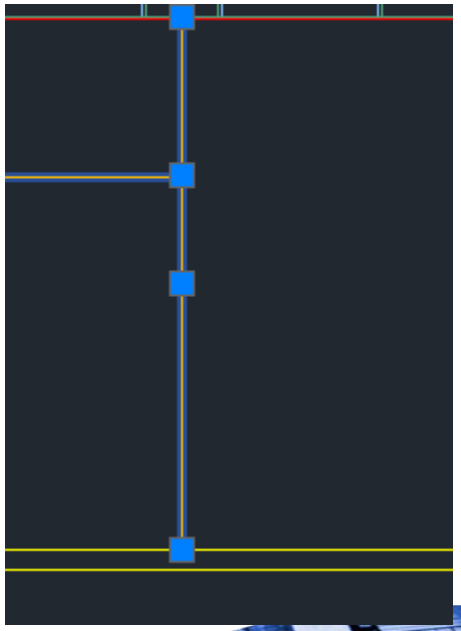
Practise with

- Fillet and chamfer
- Divide

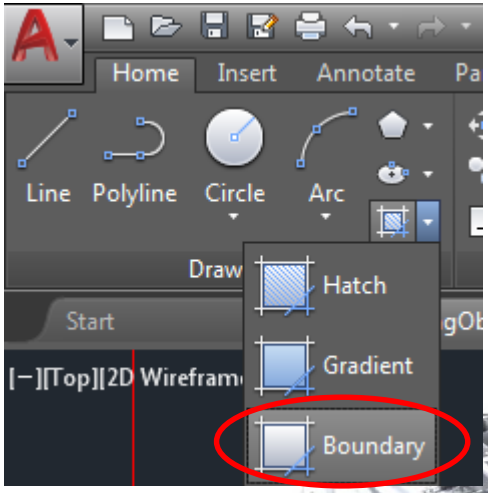


Practise with

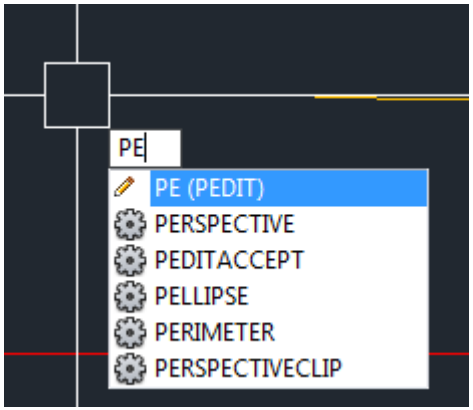
- Grips and grip editing
  - Grip: the blue block

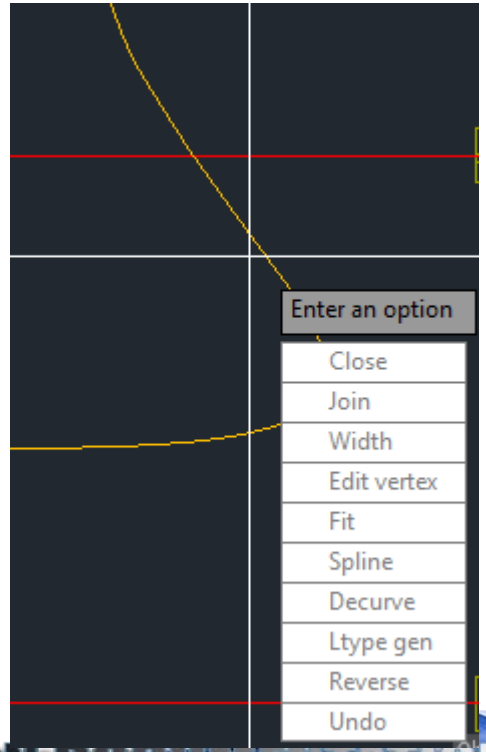


- Boundaries
  - Function: identify island for hatches



- Editing polylines and splines
  - Type ‘pe’

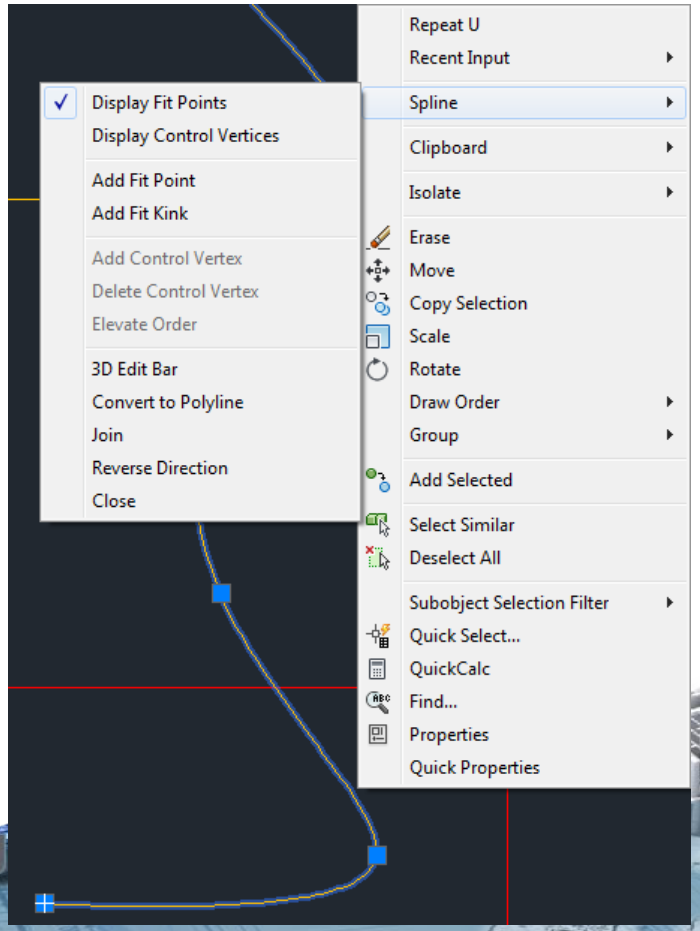




The accuracy of drawing



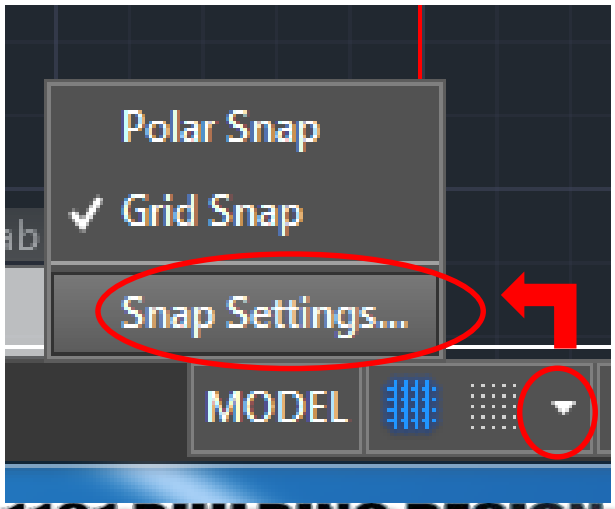
- Another way of editing splines
  - Select
  - Right click
  - Select ‘spline’

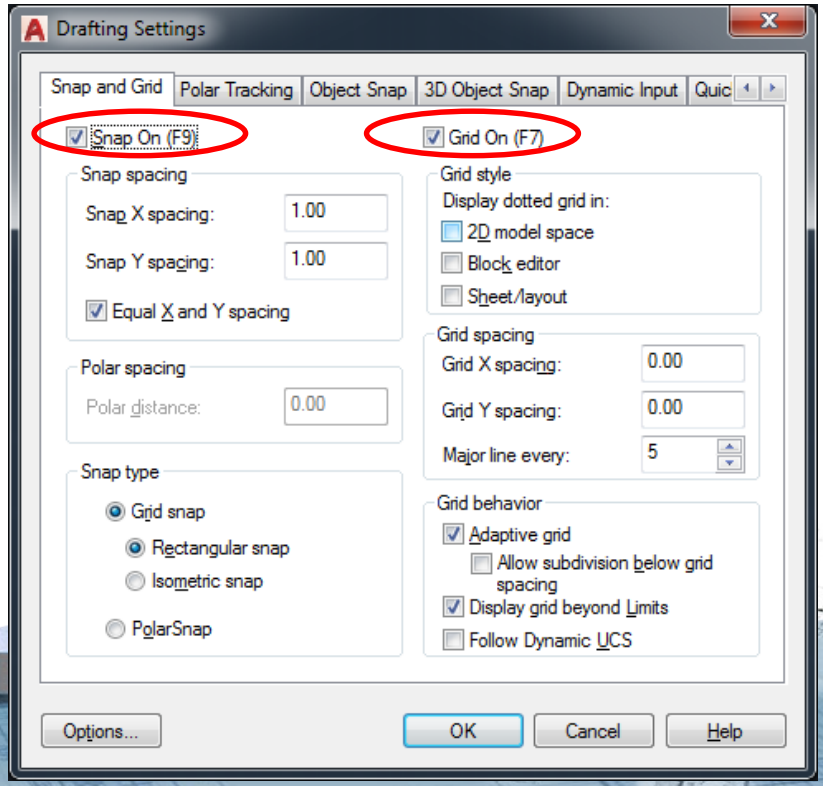


The accuracy of drawing

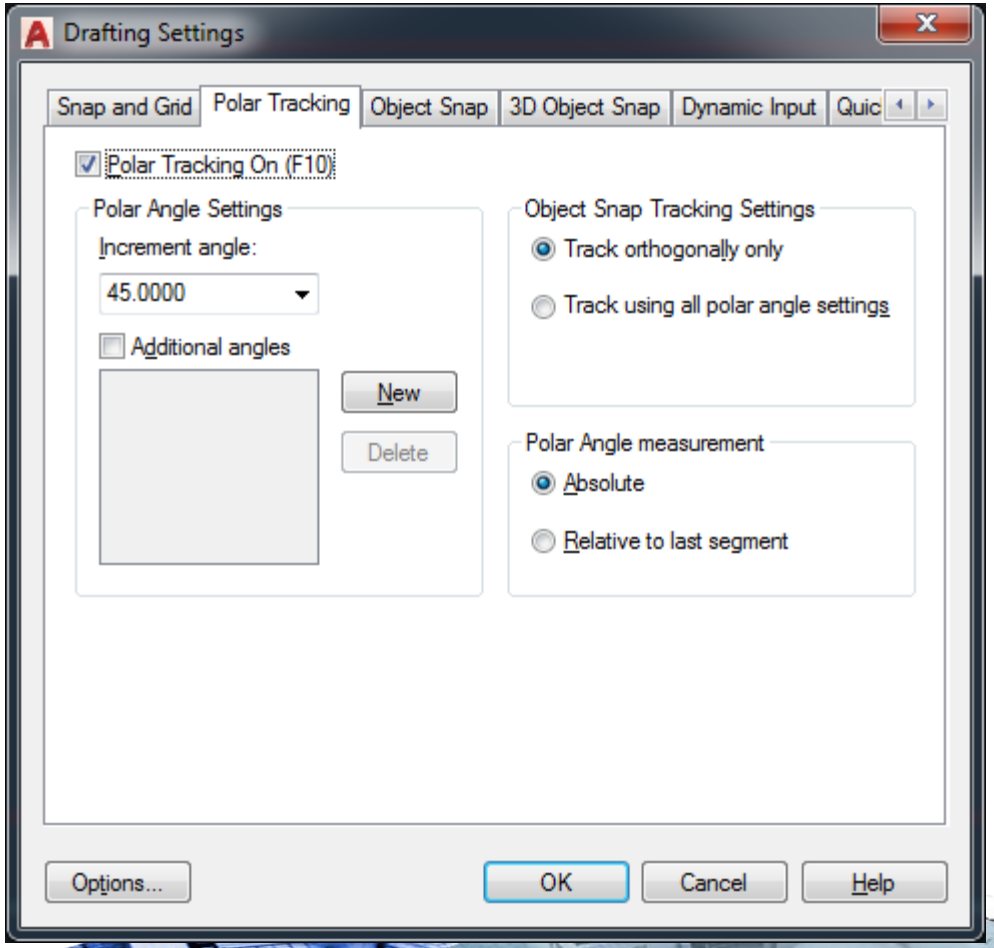
- Grid
- Snap
- Snapping settings







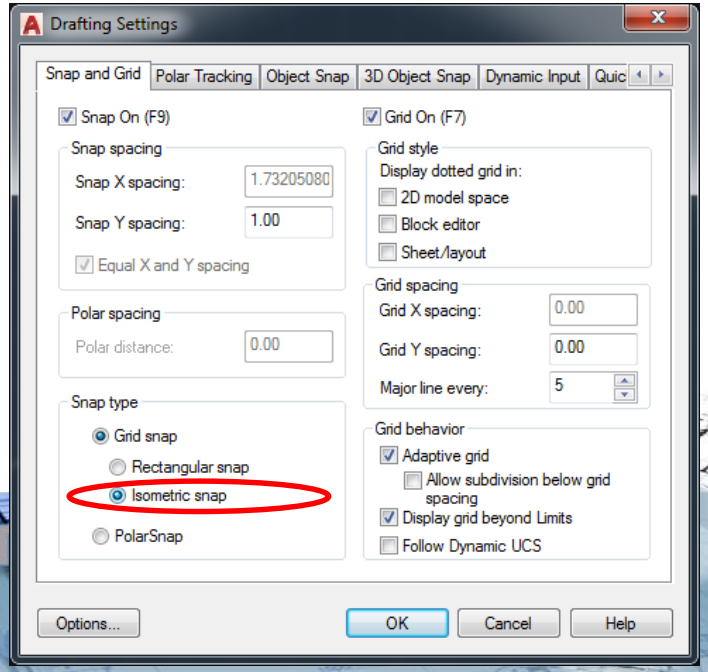
- Polar tracking
  - Track angles of specific degree



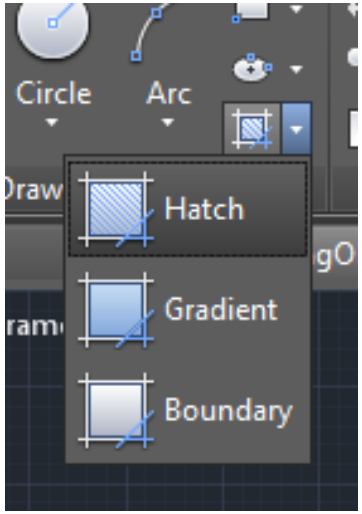
- Object snap



- Isometric snap



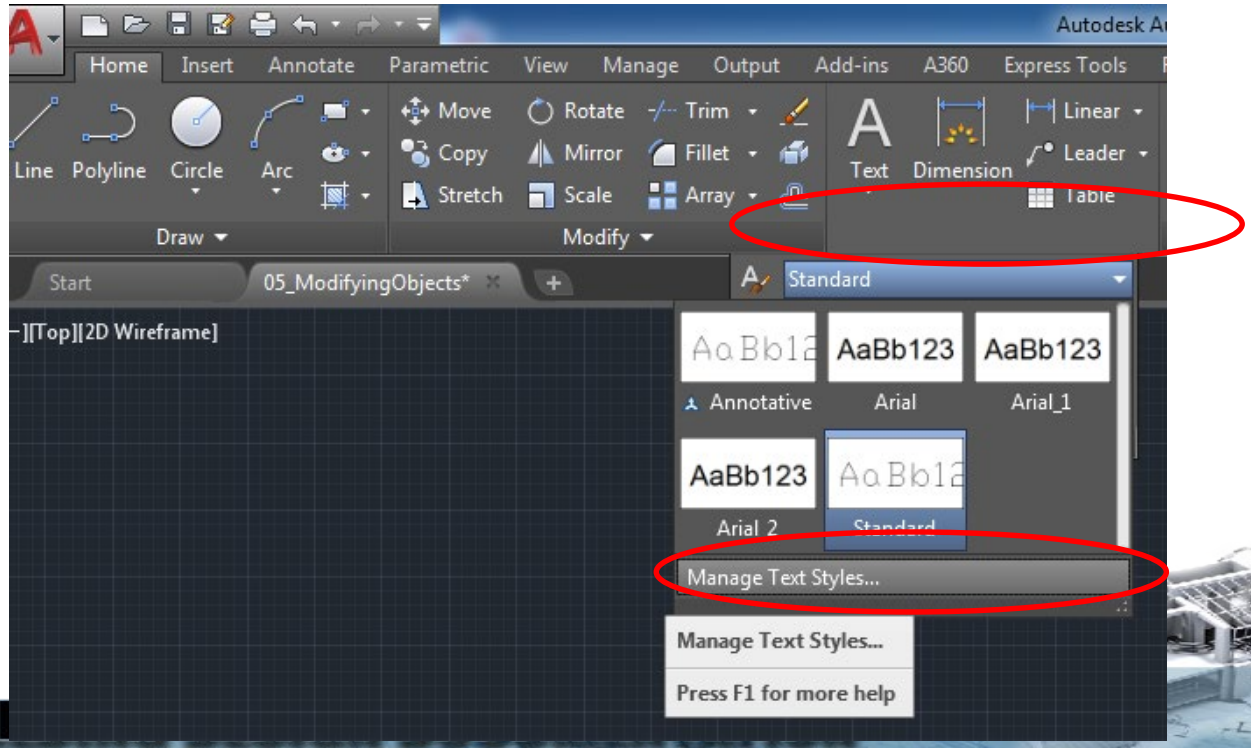
Hatch



Text



- Edit text style
  - The pull-down list of ‘annotation



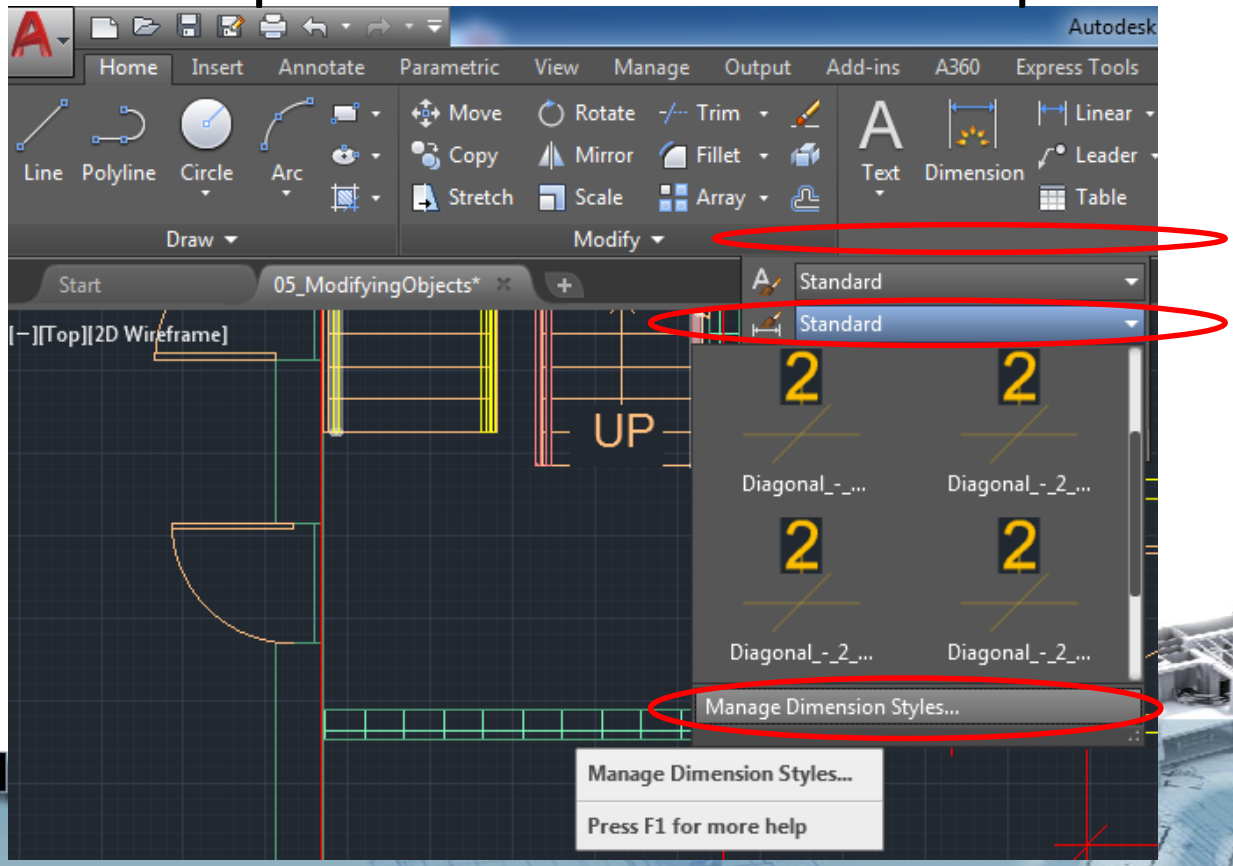
- Type in text



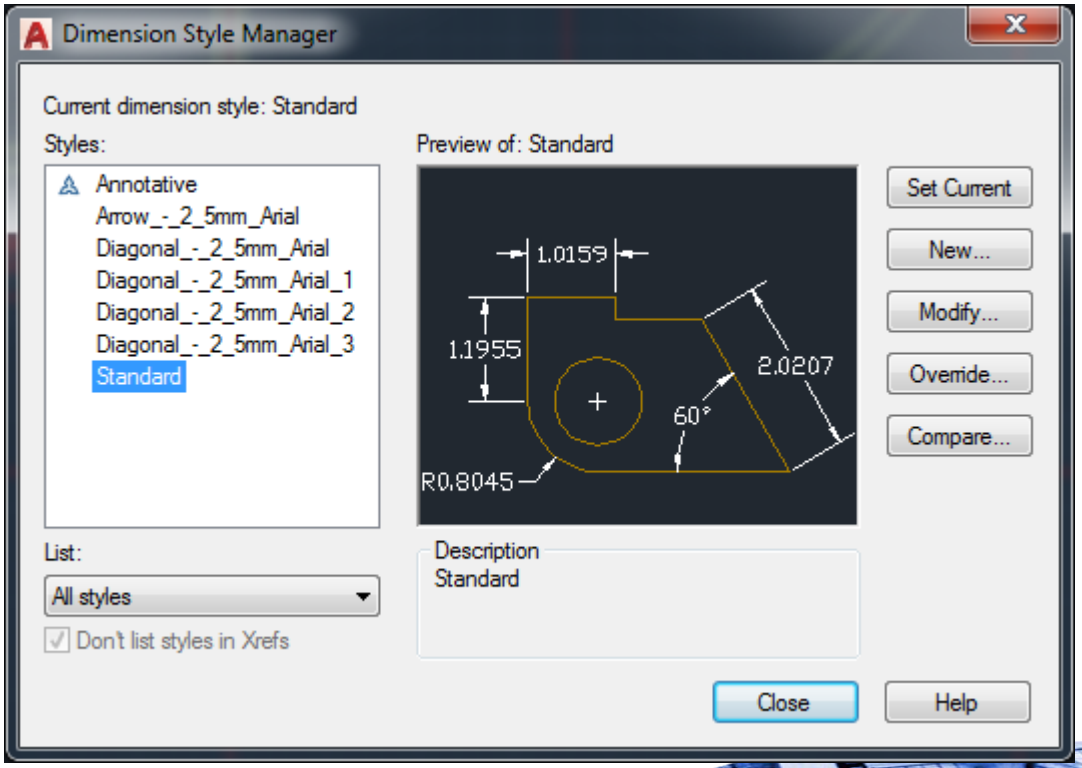
Dimensions



- Dimention styles
  - The second option of the ‘Annotation’ pull-down list

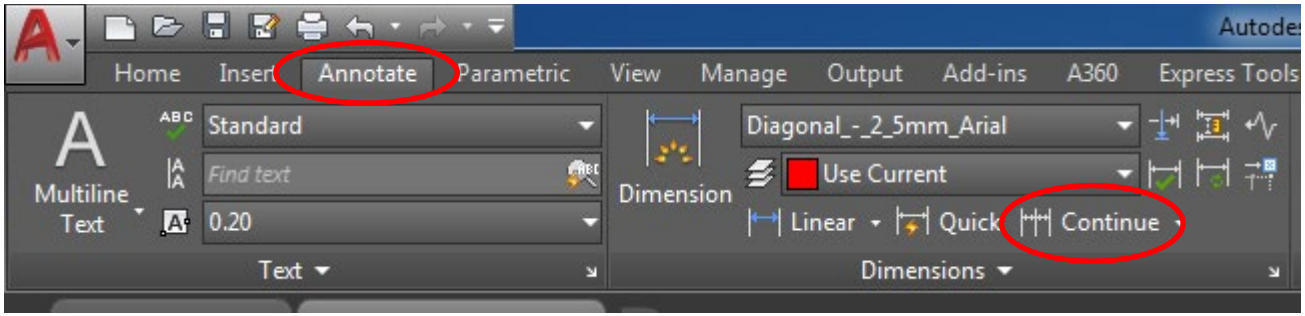


Try different settings



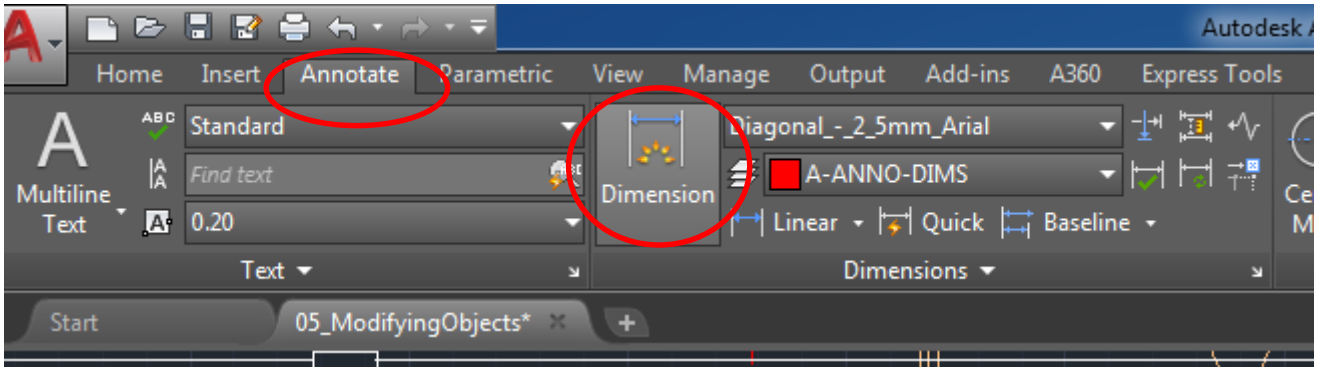
Place dimensions

- Place dimensions
  - Ensure select right **layer** and **dimension style**
  - Continue dimension



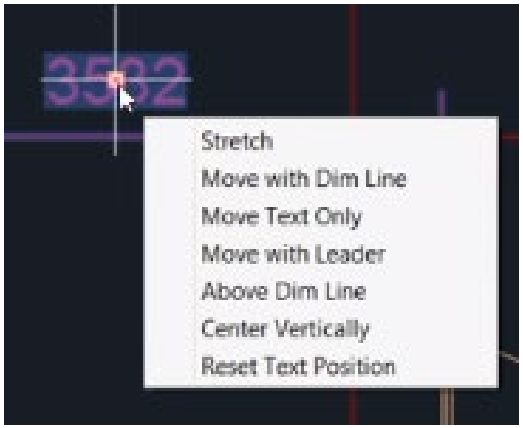
Place dimensions

- Automatic dimensioning
  - Click the object that you are going to dimension



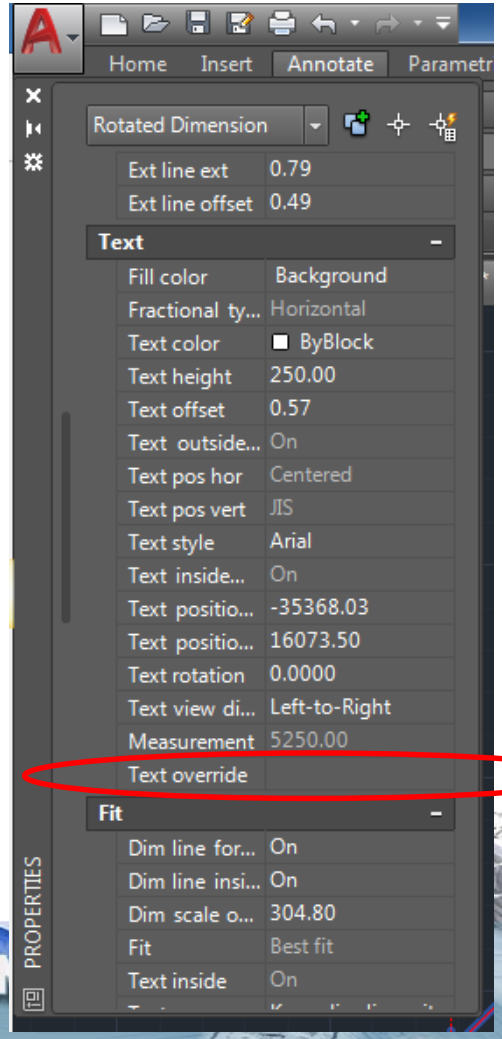
Edit dimensions

- Edit dimensions
  - Hove upon the grip(the small blue block)



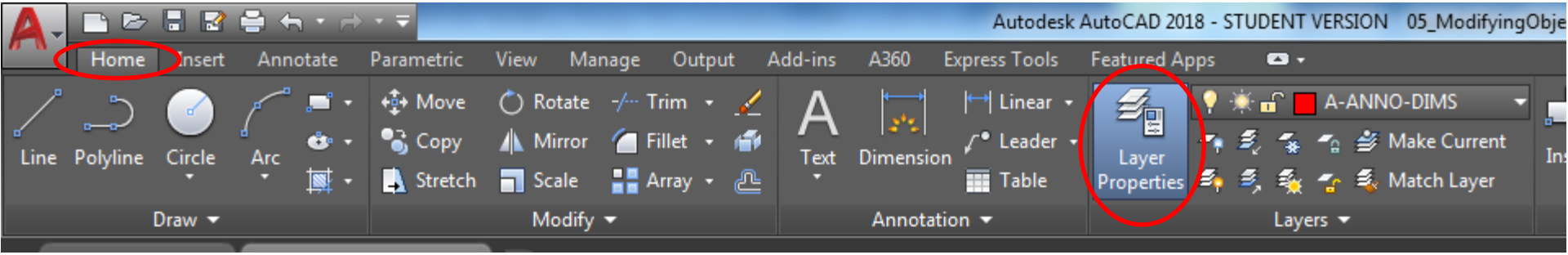
Edit dimensions

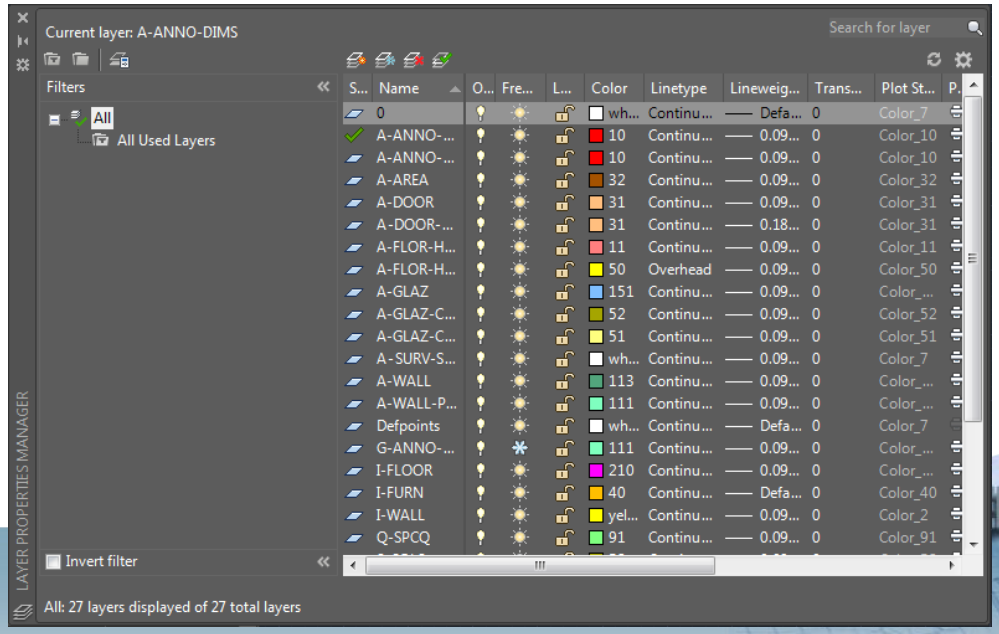
- Change the text content
  - Double click
  - OR right click, ‘properties’ and change ‘Text override’

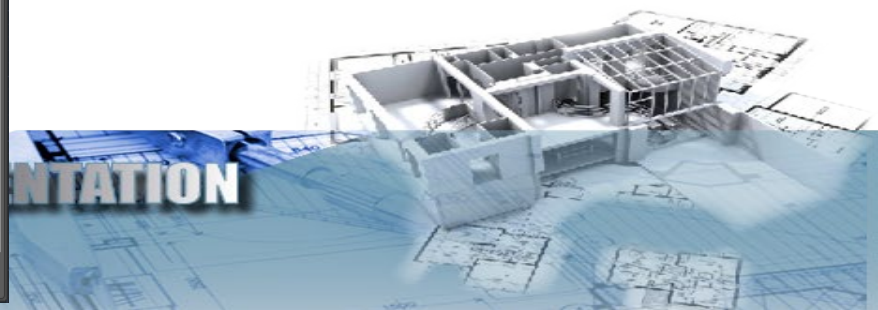


Object and layer properties

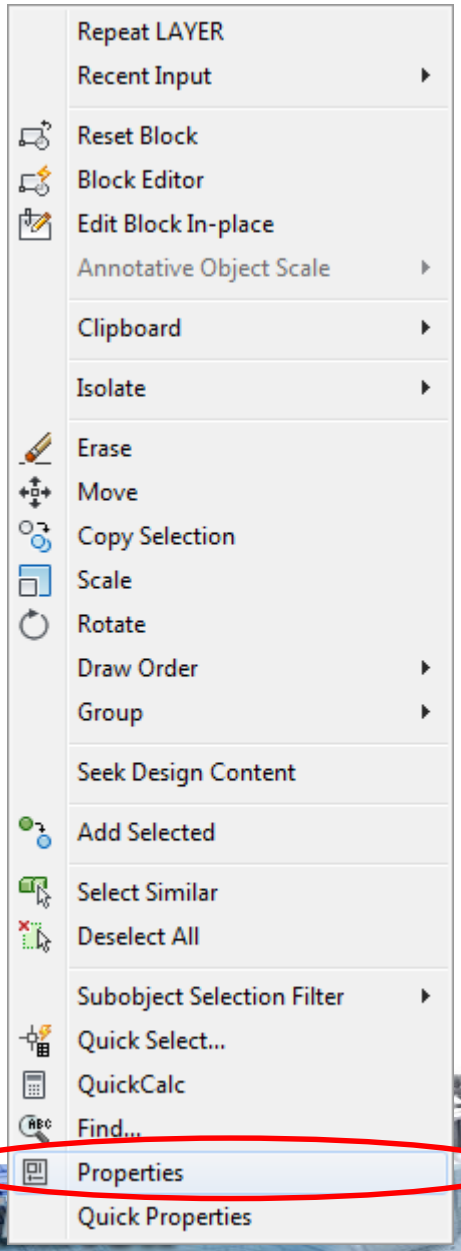
- Layer properties



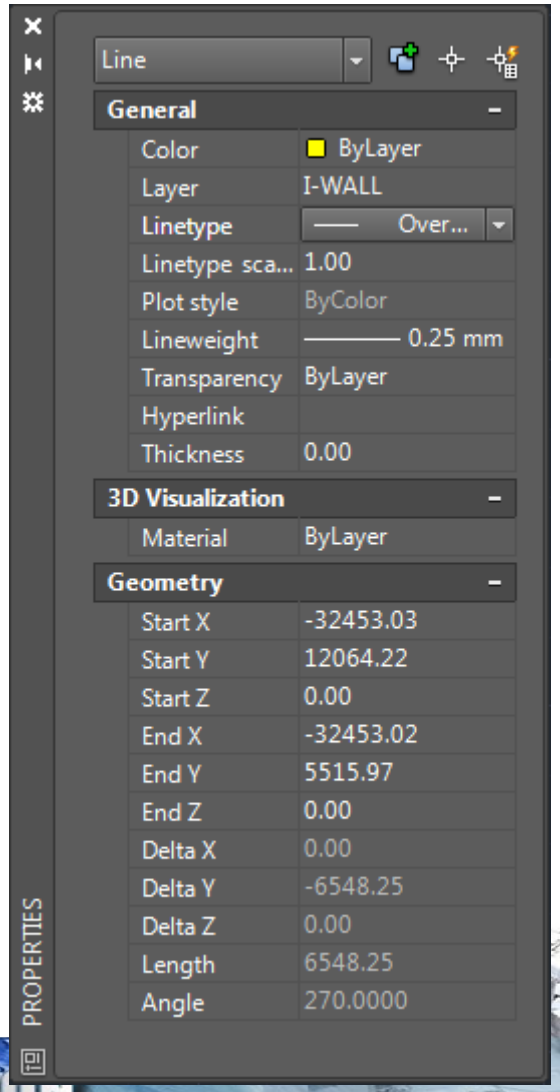




- Properties
  - Select an object
  - Right click
  - Select ‘properties



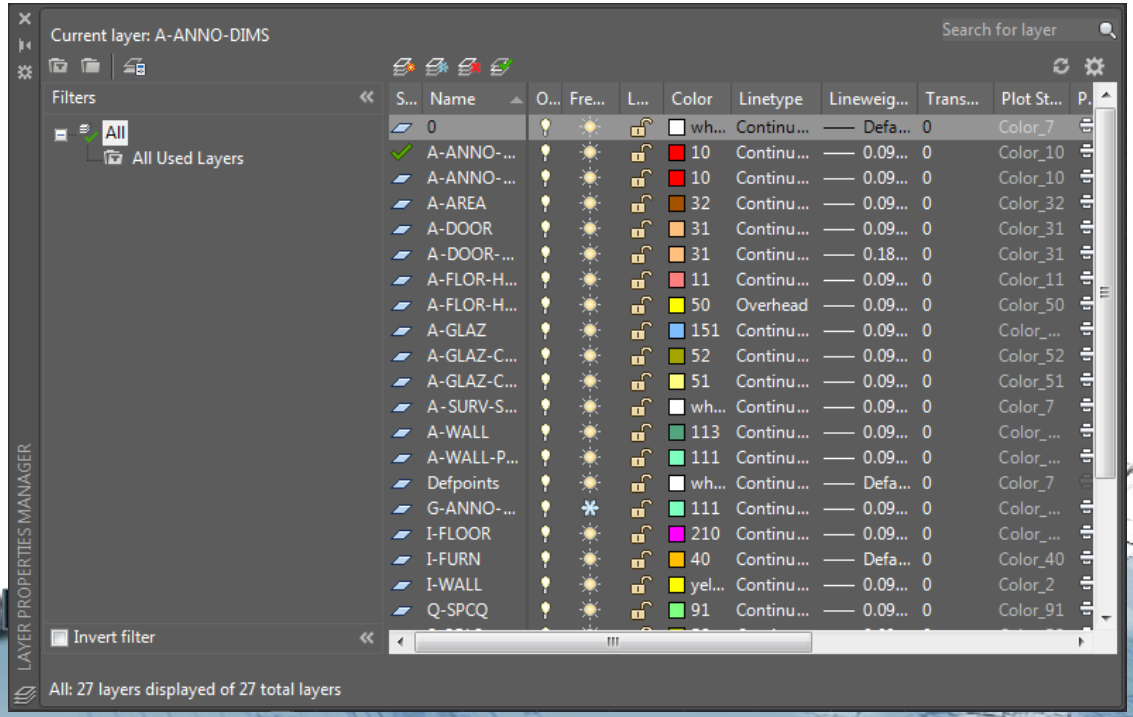
- Try to edit and modify in properties
  - Eg. Color
  - Eg. Linestye
  - …



- Ltscale
  - Select a dash
  - Type ‘LTSCALE’
  - Type a distance and observe the space change of the dash



- Create a new layer
- Change colour, linestyle…
- The use of ‘On’, ‘Freeze’ and ‘Lock’



- The co-work between ‘properties’ and ‘layer properties’
  - ‘By layer’

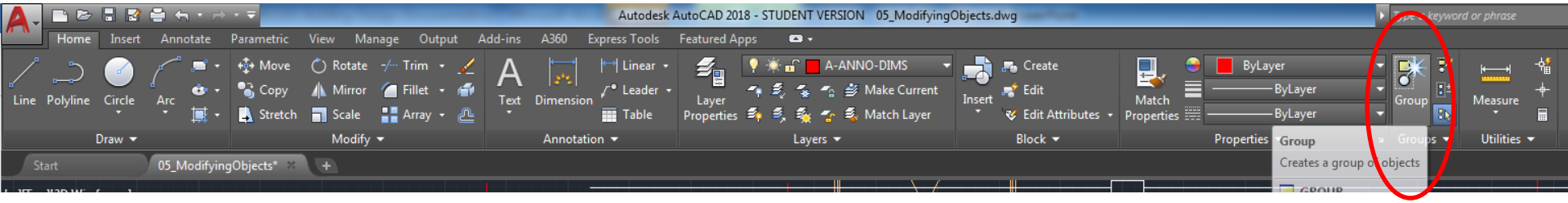


Group and block



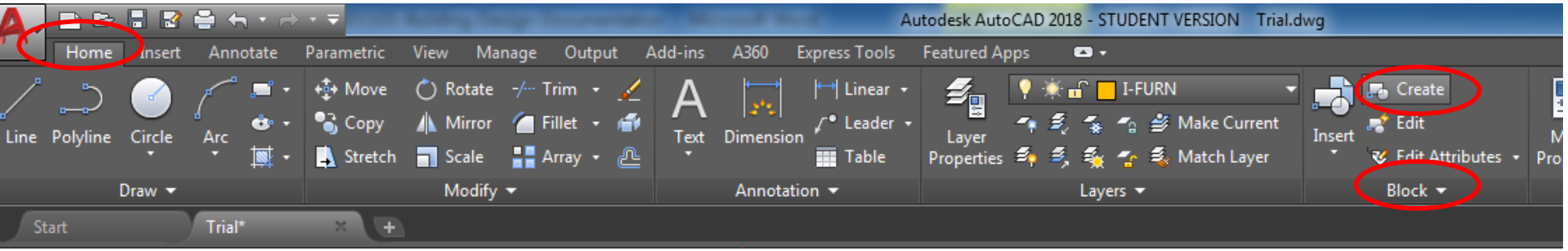
Group and block

- Create a group



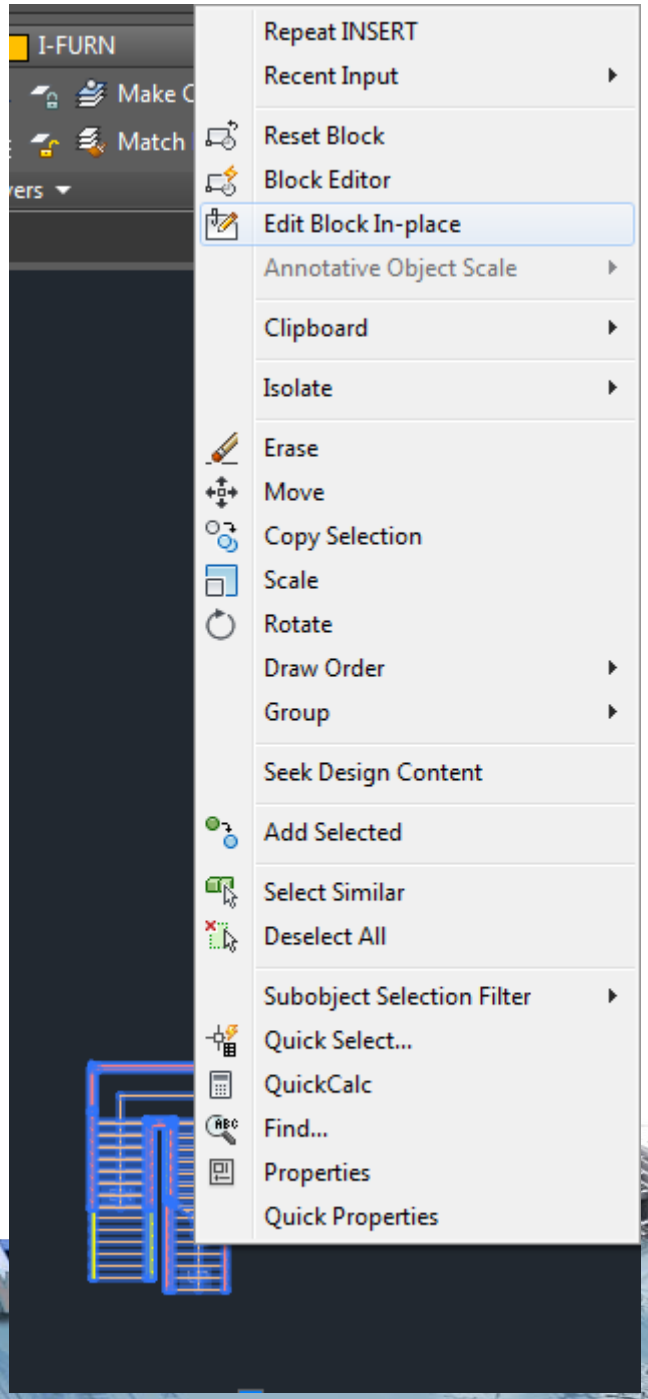
Group and block

- Create a block



- Insert a block

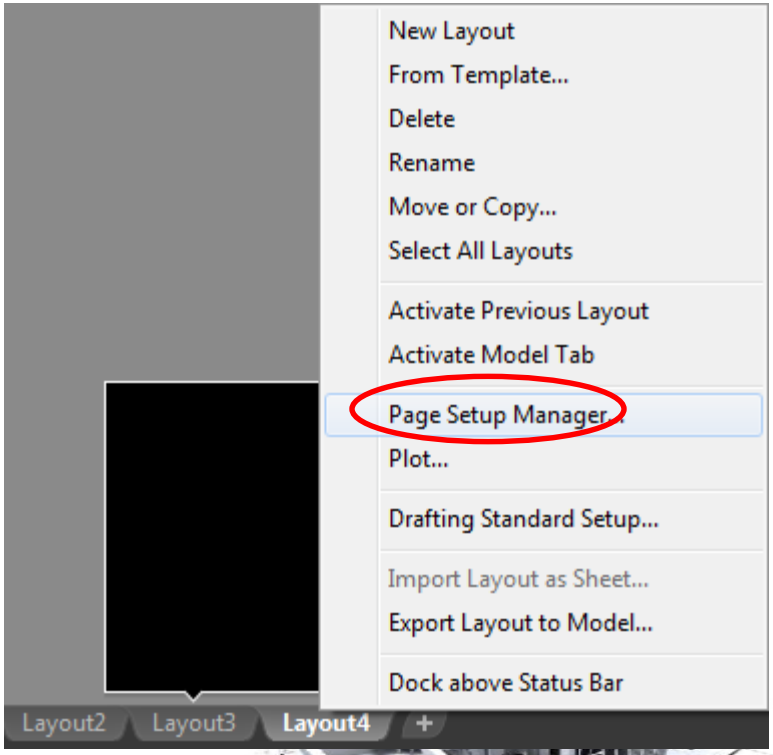
- Reedit the block
  - Select the block
  - Right click it
  - Select ‘Edit Block In-Place’
  - A global modification will be applied to all the objects of this block

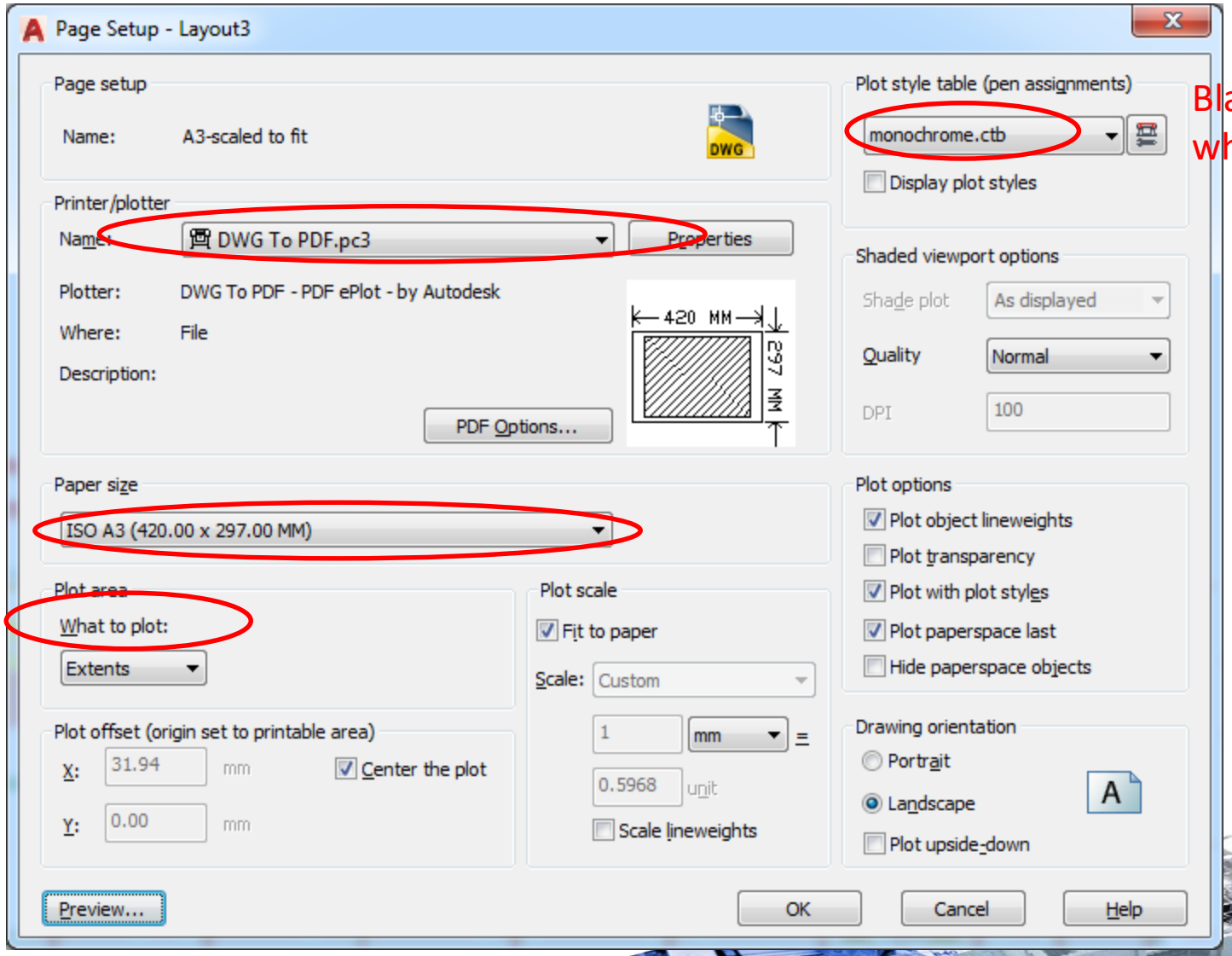


Output

Output

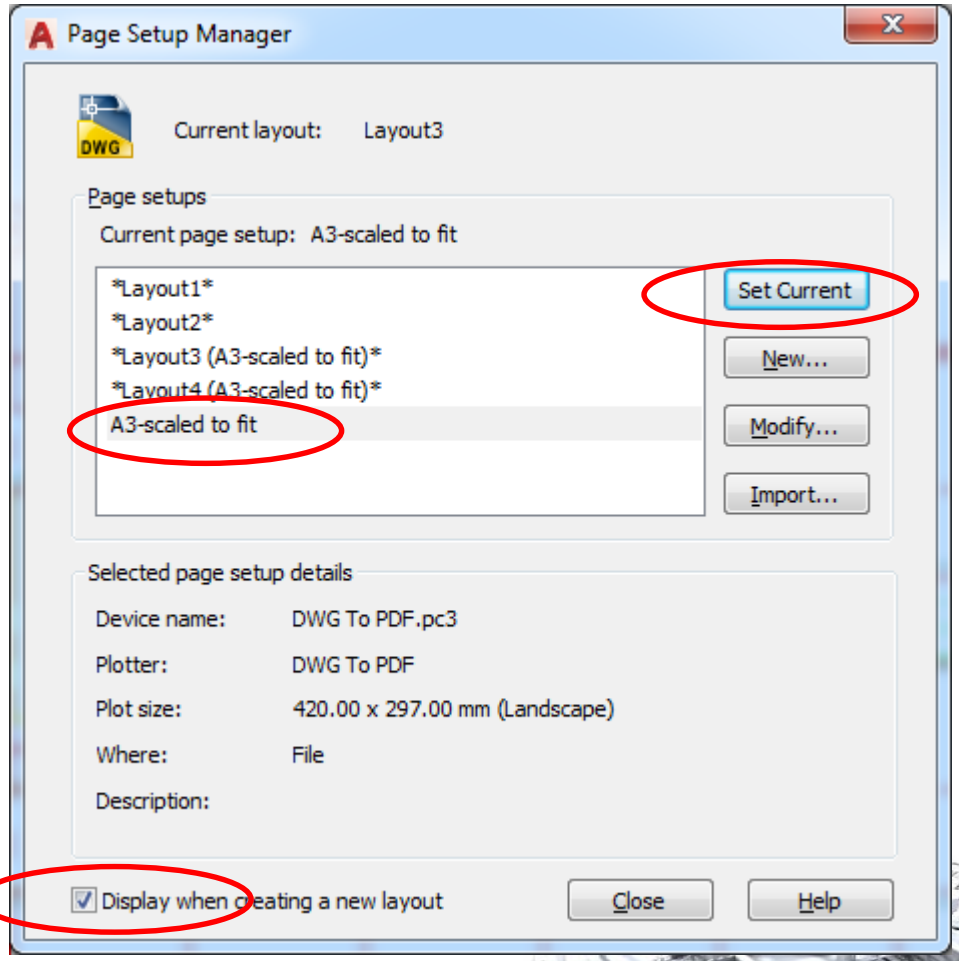
- Right click ‘Layout#’
- Select ‘Page Setup Manager’





Black and white

- Apply different page setup to the layout



- Directly export from the model
  - Right click ‘Model’
  - Select ‘Plot…’
  - Select ‘Continue to plot a single sheet’

